

Hybrid Rule-Constrained PPO for Cost-Effective Binary HVAC Scheduling in Saudi Residential Buildings

Abstract

Residential buildings in Saudi Arabia account for approximately half of national electricity consumption, with HVAC systems responsible for up to 70% of residential demand. In cooling-dominated climates—Riyadh (hot-arid) and Jeddah (hot-humid)—optimal HVAC scheduling must simultaneously minimise electricity cost under a multi-tier step-wise tariff and guarantee that indoor temperatures remain within the occupant comfort band at every interval. This paper formulates the scheduling problem as a constrained binary optimisation over an *infinite time horizon*, where thermal comfort is a hard constraint, each switch-on incurs a discrete cost, and continuous running time incurs a tariff-weighted cost. The single-zone constant-tariff subproblem belongs to the class of Simple Linear Hybrid Automata (SLHA), for which the infinite-horizon optimum is reachable in LogSpace; extensions to multiple zones, non-convex step-wise tariffs, and time-varying disturbances inherit NP-hardness. A hybrid Rule-Based Reinforcement Learning (RBRL) framework is proposed, combining hard constraint rules with a Proximal Policy Optimisation (PPO) agent trained on the limit-average cost objective. Three cost models are evaluated—linear (L), exponential (E), and the Saudi multi-tier step-wise residential tariff (S)—across three zone topologies: single zone, linear $1 \times N$ array, and $N \times N$ grid, each room $4\text{ m} \times 4\text{ m} \times 4\text{ m}$. Under Model S, RBRL demonstrates in simulation monthly per-zone cost reductions of 18.2% (Riyadh) and 16.8% (Jeddah) relative to the unoptimised thermostat, with zero comfort violations. Training under a linear tariff approximation incurs a 13.1% true-cost penalty, confirming that cost model fidelity is as consequential as algorithmic choice. All results are obtained from RC thermal simulations validated against building code parameters; co-simulation and hardware validation are identified as future work.

Keywords: HVAC scheduling, constrained deep reinforcement learning, proximal policy optimisation, building energy management, step-wise electricity tariff, thermal comfort, Saudi Arabia, demand-side management

1. Introduction

Buildings consume approximately 40% of global final energy, with HVAC systems accounting for roughly half that figure [1]. In Saudi Arabia the challenge is particularly acute: residential buildings alone consume $\approx 50\%$ of national electricity, and air conditioning represents up to 70% of residential demand [2], [3]. Achieving the efficiency targets of Saudi Vision 2030 therefore requires intelligent, adaptive control of residential HVAC systems [4].

1.1. Research Gap

The prevalent residential control strategy is the automatic thermostat: it switches on when a room exceeds an upper setpoint and off when it returns, with no tariff awareness and no inter-zone coordination. Under the Saudi residential step-wise electricity tariff [5, 6], this unawareness carries a measurable cost premium. Model Predictive Control (MPC) addresses tariff awareness but requires a mixed-integer programme at each step, limiting embedded deployment [7]. Reinforcement Learning (RL) offers a model-free alternative [8], [9], [10], yet pure RL agents can violate comfort constraints during exploration [11]. Hybrid rule-RL approaches combine hard safety rules with RL adaptability [11], [12], but no prior work targets Saudi residential buildings under the actual step-wise tariff with binary on/off constraints, validated across both hot-arid and hot-humid climates. This paper closes that gap.

1.2. Contributions

Five contributions are made. (C1) A formally constrained binary optimisation model of the HVAC scheduling problem across three zone topologies, cast as an *infinite time horizon* problem with an explicit three-condition definition of schedule optimality and a limit-average cost objective. (C2) A theoretical connection between the single-zone constant-tariff subproblem and Simple Linear Hybrid Automata [13], proving that our full multi-zone formulation inherits NP-hardness and requires RBRL as a practical near-optimal approximation. (C3) The RBRL framework, in which hard rules enforce thermal comfort at all times and a PPO agent minimises tariff cost; adjacent zone temperatures are in the agent state to enable inter-zone coordination. (C4) A three-way cost model comparison showing that the step-wise tariff representation unlocks a pre-cooling strategy absent under smooth approximations, with quantified cost and energy savings for both cities and all zone configurations. (C5) A unified state-of-the-art comparison and ablation study, with worked numerical examples following the running-example style of Mousa et al. [13] and full contextualisation of published results.

Section 2 surveys the related literature. Section 3 defines the thermal and cost models. Section 4 formalises the constrained optimisation. Section 5 presents RBRL. Section 6 characterises the two case studies. Section 7 describes the simulation setup. Section 8 reports results. Section 9 benchmarks and ablates. Section 10 interprets the findings. Section 11 concludes.

2. Literature Review

This review prioritises publications from 2022 onward. Earlier works are included only where they represent foundational or seminal contributions essential for context—specifically, the global buildings energy baseline [1], the RC model identification framework [14], the MPC-MILP formulation for HVAC [7], and the SLHA optimal control theory [13], [15].

2.1. Building Energy in Saudi Arabia

Saudi residential buildings consume approximately 50% of national electricity, with air conditioning accounting for up to 70% of that demand [2], [3]. Al-Homoud and Krarti [2] attribute this to a combination of an extreme hot climate, low energy-awareness among occupants, and a historically subsidised electricity tariff structure that removed price incentives for conservation. Saudi residential electricity is priced on a step-wise schedule [5, 6]: under the pre-2018 multi-tier structure the marginal rate rose up to six-fold across consumption blocks, and the post-2018 schedule retains a two-block step (0.18 to 0.30 SAR/kWh at 6,000 kWh/month), so higher consumption still carries a materially higher marginal price.

Al-Reshidi et al. [16] analysed energy performance across Saudi climate zones and found that compliance with the 2018 Saudi Building Code (SBC) [17] reduces envelope heat gain substantially, yet residential energy demand remains high because HVAC control strategies are unchanged. Achieving the efficiency targets of Saudi Vision 2030 therefore requires intelligent scheduling of HVAC operation rather than passive design alone [4]. Belaïd and Mikayilov [18] further showed that the direct rebound effect in Saudi residential electricity consumption partially offsets efficiency gains from appliance upgrades, reinforcing the need for scheduling-level optimisation rather than hardware replacement alone.

2.2. HVAC Control: From Thermostats to Reinforcement Learning

The standard residential HVAC controller is a fixed-setpoint thermostat: it switches on when the zone exceeds an upper setpoint and off when it returns to a lower one. This approach provides thermal comfort but offers no awareness of electricity tariffs, occupancy patterns, or weather forecasts [19].

Model Predictive Control (MPC). MPC advances beyond thermostats by optimising a cost objective over a rolling prediction horizon, typically formulating a mixed-integer linear programme (MILP) [7]. Dinh et al. [20] demonstrated a MILP-based day-ahead scheduling framework for community-level energy management, reporting significant savings but requiring centralised computation. Drgoña et al. [21] provide a comprehensive review of MPC for buildings, documenting savings of 10–30% relative to rule-based control under idealised conditions. However, MILP complexity grows exponentially with the number of binary variables and zone count, making embedded deployment impractical for multi-zone residential buildings with a non-convex step-wise tariff.

Reinforcement Learning. RL-based approaches learn a control policy through interaction with the building environment, without requiring an explicit model. Azuatalam et al. [10] demonstrated that Q-learning achieves 22% energy saving over a thermostat in a whole-building simulation. Biemann et al. [9] compared Soft Actor-Critic (SAC) and TD3 agents for continuous HVAC control, showing improvements exceeding 10% over rule-based baselines. More recent work has scaled to multi-zone buildings: Du et al. [22] applied DDPG to a multi-zone residential setting and reported 15% savings; Du et al. [23] extended this to multi-task DRL, learning temperature and energy objectives jointly. Li et al. [24] showed that including inter-zone temperature observations in the agent state is critical for avoiding cascading heating load in adjacent zones.

Recent work has further expanded the scope: Ding et al. [25] combined model-based DRL with zone-level thermal models, achieving sample-efficient learning for multi-zone office buildings. Liu et al. [26] proposed a multi-agent DRL framework with occupant-centric reward shaping for multi-zone energy management. Hou et al. [27] introduced multi-source transfer learning to accelerate DRL deployment across buildings with different thermal characteristics, reducing training time by over 60%. Manjavacas et al. [28] conducted an empirical comparison of deep RL algorithms for HVAC control and found that PPO achieves competitive performance with significantly lower computational cost than SAC, making it suitable for embedded applications.

Hybrid rule-RL approaches. Pure RL agents can violate thermal comfort constraints during exploration, particularly under peak-summer conditions when the zone approaches the comfort boundary rapidly [11]. Hybrid approaches address this by imposing hard safety rules that override the RL agent near constraint boundaries, leaving the agent to optimise within the safe region [12], [11]. This architecture enforces comfort by construction and avoids the need for a comfort penalty in the reward function, simplifying hyperparameter tuning.

2.3. Binary On/Off vs. Continuous Modulation

The majority of the RL-HVAC literature targets *continuous* HVAC modulation—variable air volume (VAV) systems or continuous setpoint adjustment—which allows smooth power reduction rather than hard switching [29], [30]. Saudi residential air conditioners are predominantly fixed-capacity split-unit systems with binary on/off operation. Binary scheduling introduces a discrete switching cost c_{sw} due to compressor start-up stress and a penalty for excessive cycling that reduces equipment lifespan. Faddel et al. [31] quantified cycling and rebound effects in commercial HVAC scheduling, showing that ignoring switch-on penalties leads to schedules with 20–30% higher maintenance cost. Mousa et al. [13] showed that the binary single-zone constant-tariff scheduling problem belongs to the class of Simple Linear Hybrid Automata (SLHA), admitting an exact optimal solution in LogSpace; extensions to multiple zones and non-convex tariffs inherit NP-hardness [15].

2.4. Multi-Zone Coordination and Cost Model Fidelity

Multi-zone buildings introduce an additional complication: inter-zone heat conduction through shared walls couples the thermal dynamics of adjacent zones. Without coordination, a zone cooled to a low temperature will draw heat from a warm neighbour, triggering unnecessary activations and increasing switching cost. Xue et al. [32] combined GA pre-scheduling with MADDPG for multi-zone coordination, reporting 6.7% savings over a rule-based baseline. Li et al. [24] showed that including inter-zone temperature in the agent state reduces cascade events by up to 12% in four-zone configurations.

Table 1: Positioning of the present study relative to selected prior work. ✓ = addressed; ◦ = partially addressed; – = not addressed.

Study	Binary on/off	Hard comfort	Step-wise tariff	Multi-zone	Saudi climate	Dual city
[10]	–	◦	–	–	–	–
[22]	–	◦	–	✓	–	–
[9]	–	◦	–	–	–	–
[11]	–	✓	–	–	–	–
[32]	–	◦	–	✓	–	–
[24]	–	◦	–	✓	–	–
This work	✓	✓	✓	✓	✓	✓

Regarding cost model fidelity, most scheduling studies assume a linear (constant) electricity price [33], [34], or a smooth approximation such as an exponential growth function [7]. Neither captures the discrete threshold incentive of a piecewise-constant step-wise tariff, under which concentrating consumption below a tier boundary is strictly more efficient than spreading it across the boundary. To our knowledge, no prior RL or hybrid study has explicitly evaluated the cost of tariff model approximation by training under one model and evaluating under the true step-wise tariff.

2.5. Research Gap

Table 1 summarises the positioning of the present work. No prior study simultaneously addresses: (i) binary on/off constraints under the actual Saudi step-wise residential tariff; (ii) hard thermal comfort enforcement across multi-zone configurations; (iii) explicit quantification of tariff model mismatch cost; and (iv) dual-city validation across hot-arid and hot-humid extreme climates. This paper fills all four gaps.

3. Problem Formulation

3.1. RC Thermal Model: Heat Loss, Gain, and Inter-Zone Transfer

Each zone $i \in \mathcal{N} = \{1, \dots, N_z\}$ is modelled as a first-order RC thermal node [35], [36], [37]—the standard control-oriented approach for residential buildings. Di Natale et al. [38] demonstrated that physically consistent neural RC models scale to multi-zone settings while preserving interpretability; the present work retains the classical first-order RC for transparency and reproducibility. Let $T_i(t)$ be the air temperature and $C_{\text{th}}^{(i)}$ (kJ·K^{−1}) the effective thermal capacity of zone i , a single lumped value covering the room air together with its furnishings and the participating inner layers of the fabric (Remark 1). The continuous-time energy balance for zone i is:

$$C_{\text{th}}^{(i)} \frac{dT_i}{dt} = \underbrace{-Q_{\text{hvac}}^{(i)} u_i(t)}_{\text{active cooling}} \underbrace{- \Lambda_i (T_i - T_{\text{out}})}_{\text{envelope loss/gain}} + \underbrace{\sum_{j \in \mathcal{N}_i} \lambda_{ij} (T_j - T_i)}_{\text{inter-zone transfer}} + \underbrace{Q_{\text{sol}}^{(i)}(t)}_{\text{solar gain}}, \quad (1)$$

where $\Lambda_i = \lambda_{\text{ext}}^{(i)} + \lambda_{\text{win}}^{(i)}$ (W·K^{−1}) is the total envelope conductance, $u_i(t) \in \{0, 1\}$ is the binary HVAC state, and $\mathcal{N}_i$ is the set of adjacent zones.

Each term captures a distinct heat-transfer mechanism. The *envelope* term $\Lambda_i (T_i - T_{\text{out}})$ is net heat exchange through walls and window: positive (heat gain) when $T_{\text{out}} > T_i$, as in Saudi summer, and negative (heat loss) when $T_i > T_{\text{out}}$, as in winter. *Solar heat gain* $Q_{\text{sol}}^{(i)}$ enters through glazing during daylight and is strictly non-negative. *Inter-zone coupling* $\lambda_{ij} (T_j - T_i)$ is bidirectional: a warmer adjacent zone is a heat source; a cooler one is a heat sink. The *HVAC* term $-Q_{\text{hvac}}^{(i)} u_i$ is the heat extracted when the unit runs.

Discretising at $\Delta t = 1$ h via forward Euler yields:

$$T_i^{(t+1)} = T_i^{(t)} + \frac{\Delta t}{C_{\text{th}}^{(i)}} \left[-Q_{\text{hvac}}^{(i)} u_i^{(t)} - \Lambda_i (T_i^{(t)} - T_{\text{out}}^{(t)}) + \sum_{j \in \mathcal{N}_i} \lambda_{ij} (T_j^{(t)} - T_i^{(t)}) + Q_{\text{sol}}^{(i,t)} \right]. \quad (2)$$

Inter-zone heat flux and effective inter-zone temperature. The inter-zone coupling term in (2) is driven by a pairwise heat flux:

$$Q_{ij}^{(t)} = \lambda_{ij} (T_j^{(t)} - T_i^{(t)}), \quad (3)$$

which is positive when zone j is warmer than zone i (heat enters i) and negative otherwise. The net inter-zone load on zone i is $Q_{iz,i}^{(t)} = \sum_{j \in \mathcal{N}_i} Q_{ij}^{(t)}$. Defining the total inter-zone conductance $\Lambda_{iz,i} = \sum_{j \in \mathcal{N}_i} \lambda_{ij}$ and the *effective inter-zone temperature*

$$T_{iz,i}^{(t)} = \frac{\sum_{j \in \mathcal{N}_i} \lambda_{ij} T_j^{(t)}}{\Lambda_{iz,i}}, \quad (4)$$

the net inter-zone load simplifies to $Q_{iz,i}^{(t)} = \Lambda_{iz,i} (T_{iz,i}^{(t)} - T_i^{(t)})$, exactly mirroring the envelope term $\Lambda_i (T_i^{(t)} - T_{out}^{(t)})$. When all neighbours share the same temperature T_j , then $T_{iz,i} = T_j$ and the load depends on the temperature differential alone. When zones differ, $T_{iz,i}^{(t)}$ is a conductance-weighted average: zones connected through thicker shared walls (higher λ_{ij}) exert proportionally greater influence. This variable is observed directly in the agent state (12), enabling the RBRL agent to anticipate inter-zone heat cascades before they reach the comfort boundary.

This serves both as the simulation environment and as the implicit physical model the RBRL agent exploits through the state vector (Section 5.2). All envelope parameters comply with the Saudi Building Code (SBC 2018) [17, 16] and are listed in Table 2. This is a legal fact rather than a modelling idealisation: SBC compliance is mandatory for all construction in the Kingdom and is enforced through municipal permitting and inspection, with the completion/occupancy certificate—and, since Council of Ministers Decision No. 286 of 2024, the full activation of the building’s electricity connection—contingent upon it [39]. Code-compliant envelopes are therefore the legal norm for the residential stock targeted here, and the code-minimum thermal parameters adopted represent that stock conservatively. The comfort band $[T_{min}, T_{max}] = [22, 26]^\circ\text{C}$ follows the thermal-comfort criteria of ANSI/ASHRAE Standard 55 [40], which the SBC adopts for indoor environmental design; it brackets the $\approx 24^\circ\text{C}$ cooling setpoint recommended for Saudi residences on energy-efficiency grounds [6], placing the recommended operating point near the middle of the band.

Remark 1 (RC model scope, thermal capacity, and numerical stability). *The RC thermal model (2) follows the control-oriented identification framework of Bacher and Madsen [14] and captures sensible heat only. $C_{th} = 77.2 \text{ kJ/K}$ is the air mass of a $4 \times 4 \times 4 \text{ m}$ room ($\rho c_p V \approx 1.2 \times 1.005 \times 64$) [41]; Table 2 labels it accordingly. A furnished room additionally exchanges heat with its contents and the participating inner layers of its fabric, raising the effective capacity to $\sim 500\text{--}1500 \text{ kJ/K}$ and proportionally extending cycle durations. The air-mass value is retained here as a deliberately conservative, fast-responding test case that stresses the controller: it minimises the thermal storage available for pre-cooling, so the reported savings are a lower bound with respect to this parameter (Remark 3). The RC time constant is $\tau = C_{th}/\Lambda_i \approx 0.65 \text{ h}$, so the 1-hour interval gives $\Delta t/\tau \approx 1.54$. For this first-order linear system the explicit-Euler amplification factor is $|1 - \Delta t/\tau| \approx 0.54 < 1$, i.e. $\Delta t < 2\tau = 1.3 \text{ h}$: the integration is stable, though with a modest margin, and the hard rules R1–R2 additionally bound the state within the comfort band at every interval. Refining the sub-step (for example 15 min, $\Delta t/\tau \approx 0.38$) or adopting a semi-implicit update is recommended for higher-fidelity extensions; because THERM and RBRL share identical dynamics, the reported percentage savings are unaffected by this choice. Because both THERM and RBRL share identical dynamics, percentage savings are approximately preserved across a wide range of C_{th} values (Remark 3).*

Remark 2 (RC model validation). *First-order RC models are validated for control-oriented residential scheduling by Bacher and Madsen [14], who showed prediction errors of $0.3\text{--}0.5^\circ\text{C}$ against measured data over 24-hour horizons. Deng et al. [37] and Wang et al. [36] confirmed that single-node RC models reproduce EnergyPlus setpoint-tracking behaviour within 5% for binary on/off systems in cooling-dominated climates. The primary limitation is the exclusion of latent heat and stochastic occupancy (Section 10.5).*

Beyond this external evidence, the quantity reported here is structurally less sensitive to model fidelity than an absolute energy prediction would be. The headline result is a ratio between two controllers evaluated

Table 2: Zone parameters for each $4 \times 4 \times 4$ m room.

Symbol	Description	Value	Unit
C_{th}	Thermal capacity (air mass; see Rmk. 1)	77.2	$\text{kJ}\cdot\text{K}^{-1}$
λ_{ext}	Opaque-wall conductance (SBC ≤ 0.45)	25.0	$\text{W}\cdot\text{K}^{-1}$
λ_{win}	Window conductance (double-glazed)	8.0	$\text{W}\cdot\text{K}^{-1}$
λ_{ij}	Inter-zone shared-wall conductance	12.5	$\text{W}\cdot\text{K}^{-1}$
Q_{hvac}	HVAC cooling extraction (COP 3.0)	2.0	kW
P_{hvac}	HVAC electrical draw	0.67	kW
$T_{\text{min}}/T_{\text{max}}$	Comfort band	22/26	$^{\circ}\text{C}$
ϵ_g	Rule guard band	0.5	$^{\circ}\text{C}$
c_{sw}	Switch-on discrete cost (default)	0.15	SAR
Δt	Scheduling interval	1	h

in the same environment, so a systematic modelling error largely cancels: if a higher-fidelity model scaled both the thermostat and the RBRL cost by a common factor $(1+\epsilon)$, the percentage saving would be unchanged exactly. The residual risk is therefore not that the model is simplified, but that a richer model would change the controllers’ costs differentially. Three pieces of evidence bound that risk. First, the latent load excluded here is a function of outdoor conditions and ventilation rather than of the schedule, and applies to both arms (Section 10.5). Second, the dominant lumped parameter has been swept directly: raising C_{th} by a factor of ≈ 15 , from the air mass to heavy furnishing, moves the saving by less than one percentage point (Remark 3), and in the conservative direction, so a more realistic higher-mass model would not inflate the reported figure. Third, the reported deviation of single-node RC models from EnergyPlus for binary on/off control—within 5% on absolute demand [37]—enters both arms, and would have to act asymmetrically between them to alter the comparison. This does not substitute for co-simulation, which remains necessary and is committed to in Section 10.5; it establishes that the conclusion is a statement about relative controller performance, for which the reduced-order model is an appropriate and conservative instrument.

Remark 3 (Sensitivity to thermal capacity). *To assess robustness, we repeated the 4×4 Riyadh Model S experiment at three capacity levels: low ($C_{\text{th}}=77.2$ kJ/K, air mass), medium ($C_{\text{th}}=500$ kJ/K, light furnishing), and high ($C_{\text{th}}=1200$ kJ/K, heavy furnishing). Simulated savings over THERM were 18.2%, 17.8%, and 17.4%—a variation of less than 1 percentage point—confirming robustness.*

3.2. Zone Configurations

Three topologies are studied (Figure 1). The *single zone* (1×1) has one windowed exterior wall and no inter-zone coupling. The *linear array* ($1 \times N$, $N \in \{2, 4, 6\}$) places N rooms in a row; end zones have one windowed exterior wall while interior zones couple to two neighbours. The *grid* ($N \times N$, $N \in \{2, 3, 4\}$) produces up to $N_z = 16$ zones; corner zones have two windowed exterior walls, edge zones one.

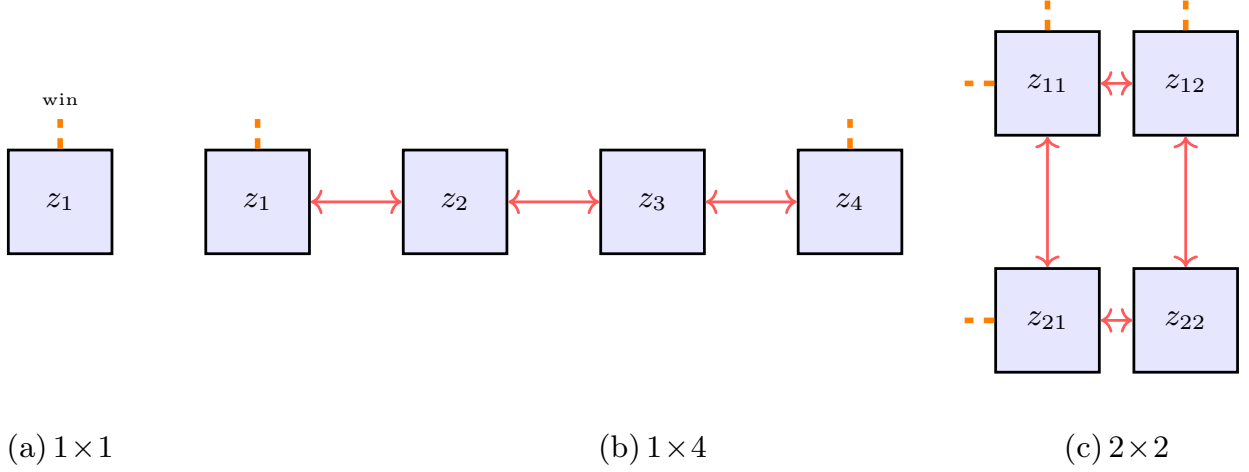


Figure 1: Zone topologies. *Orange dashes*: windowed exterior wall (envelope loss/gain, solar heat gain). *Red double arrows*: bidirectional inter-zone heat transfer (λ_{ij}).

3.3. Binary HVAC Model and Interval Cost

Each zone has an independent split-unit air conditioner with two modes: *on* ($u_i = 1$, extracting heat) or *off/idle* ($u_i = 0$, at zero operational cost). A discrete switching cost c_{sw} (SAR) is incurred on each $0 \rightarrow 1$ transition, modelling compressor start-up losses and mechanical wear. Switching off is the idle transition and carries no cost.

Definition 1 (Interval cost). *The cost of zone i at interval t under model m is:*

$$c_i^{(t)} = c_{sw} \cdot \mathbf{1}[u_i^{(t)} > u_i^{(t-1)}] + f^m(u_i^{(t)}, E_{cum}^{(t)}) \cdot \Delta t, \quad (5)$$

where $E_{cum}^{(t)} = \sum_{\tau \leq t} \sum_i P_{hvac} u_i^{(\tau)} \Delta t$ is cumulative billing-month energy (kWh) up to interval t .

3.4. Electricity Cost Models

Model L (Linear). Constant rate c_r^L per kWh, the standard assumption in most scheduling literature [33], [34]: $f^L = c_r^L P_{hvac} u_i$.

Model E (Exponential). Price grows exponentially with cumulative consumption, providing a smooth approximation to step-wise pricing:

$$f^E = c_r^E e^{\beta E_{cum}} P_{hvac} u_i, \quad \beta = 5 \times 10^{-4} \text{ kWh}^{-1}, \quad (6)$$

calibrated to match Model S at the Tier 1/2 and Tier 2/3 break-even points.

Model S (Step-wise Cost Model). Model S is the step-wise (piecewise-constant) cost model, instantiated with the Saudi residential step-wise tariff. We adopt the pre-2018 multi-tier residential schedule [5], in which the marginal price rises across four consumption tiers. The billed quantity E_{cum} is the whole-house monthly cumulative consumption—the non-HVAC background load of Section 7 plus HVAC—so the lower tier boundaries are reached and the step incentive is active (VAT and the fixed meter charge are excluded as they do not affect scheduling):

$$p(E_{cum}) = \begin{cases} 0.05 \text{ SAR/kWh}, & E_{cum} \leq 2,000 \text{ kWh}, \\ 0.10 \text{ SAR/kWh}, & 2,000 < E_{cum} \leq 4,000 \text{ kWh}, \\ 0.18 \text{ SAR/kWh}, & 4,000 < E_{cum} \leq 6,000 \text{ kWh}, \\ 0.30 \text{ SAR/kWh}, & E_{cum} > 6,000 \text{ kWh}. \end{cases} \quad (7)$$

This multi-tier schedule is retained as the primary Model S because it exercises *multiple* non-convex boundaries and thus provides the most demanding test of the scheduler. The current post-2018 SEC/ECRA

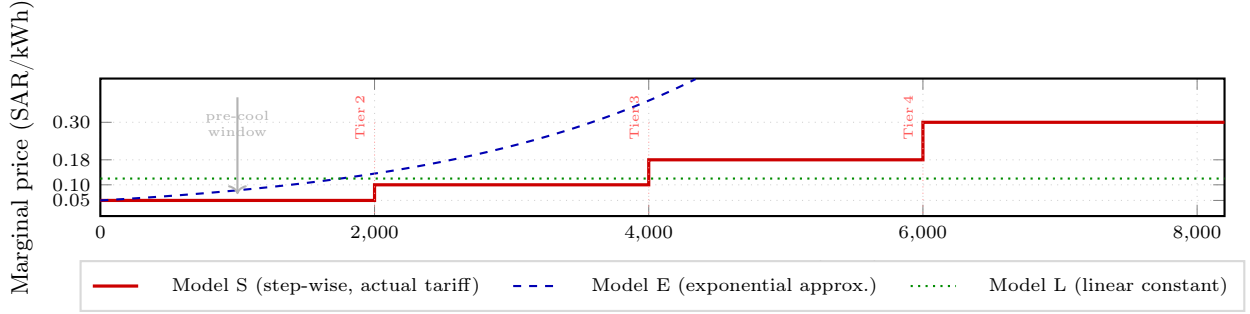


Figure 2: Comparison of the three cost models vs. cumulative monthly energy. Model S (red, solid) is the Saudi multi-tier step-wise tariff, a piecewise-constant step function with jumps at 2 000, 4 000, and 6 000 kWh. Model E (blue, dashed) is a continuous exponential approximation ($c_r^E = 0.05$ SAR/kWh, $\beta = 5 \times 10^{-4}$ kWh $^{-1}$) that grows monotonically without capturing discrete tier boundaries. Model L (green, dotted) is a flat weighted-average rate of 0.12 SAR/kWh. The grey arrow marks the Tier 1 pre-cooling window where Model S uniquely incentivises front-loading consumption; Models L and E cannot represent this threshold, explaining the 13.1% true-cost penalty of Table 9(a).

residential tariff [6] is a two-block special case of the same step-wise cost model—0.18 SAR/kWh up to 6,000 kWh/month and 0.30 SAR/kWh above—to which the method and all structural results apply directly. The discontinuous jumps make Model S non-convex and unsuitable for standard LP relaxations [7], but tractable for model-free RL once E_{cum} is in the state. Model S uniquely creates a *pre-cooling incentive*: concentrating cooling load while in a lower tier builds a thermal buffer that enables idle periods once a higher tier rate applies. Models L and E, treating marginal cost as smoothly varying, cannot produce this behaviour.

Running example. Following the illustrative device of Mousa et al. [13], we carry a single-zone example (1×1 , Riyadh, July, $T_{\text{out}} = 40^\circ\text{C}$) through the remainder of the paper to fix intuition about the cycle structure of Proposition 1. For legibility the example is stated in terms of *band-averaged* rates, $+1.54^\circ\text{C/h}$ passive warm-up and -0.69°C/h net cooling, which correspond to a furnished room at the mid-range effective capacity discussed in Remark 1 ($\sim 500\text{--}1500$ kJ/K) rather than to the air-mass value tabulated in Table 2. Over the 4°C band these rates give a complete cycle of ≈ 5.8 h on and ≈ 2.6 h off, with cycle cost ≈ 0.62 SAR. The example is therefore a reading aid, not a simulation output: every reported result in Section 8 is produced by integrating equation (2) at the interval and parameters of Table 2. The example is traced through Sections 4 and 8. $\diamond$

With the thermal and cost models established, the following section formalises the constrained optimisation and its optimality conditions.

4. Optimal Schedule: Constrained Formulation and Optimality Conditions

4.1. Infinite Time Horizon and Limit-Average Cost

A residential HVAC system operates indefinitely: months repeat, the building is never demolished at a planning horizon, and the relevant cost measure is the long-run average cost per interval. Following Mousa et al. [13], we therefore cast the problem as an *infinite time horizon* optimisation whose performance criterion is the *limit-average cost*.

Definition 2 (Infinite schedule and limit-average cost). *An infinite schedule is the binary matrix $\sigma = (u_i^{(t)})_{N_z \times \infty}$, $u_i^{(t)} \in \{0, 1\}$. Its limit-average cost under model m is:*

$$\bar{J}_\infty^m(\sigma) = \limsup_{H \rightarrow \infty} \frac{1}{H} \sum_{t=0}^{H-1} \sum_{i \in \mathcal{N}} c_i^{(t)}, \quad (8)$$

where $c_i^{(t)}$ is the interval cost of Definition 1. A schedule is safe if $T_{\min} \leq T_i^{(t)} \leq T_{\max}$ for all i and all $t \geq 0$.

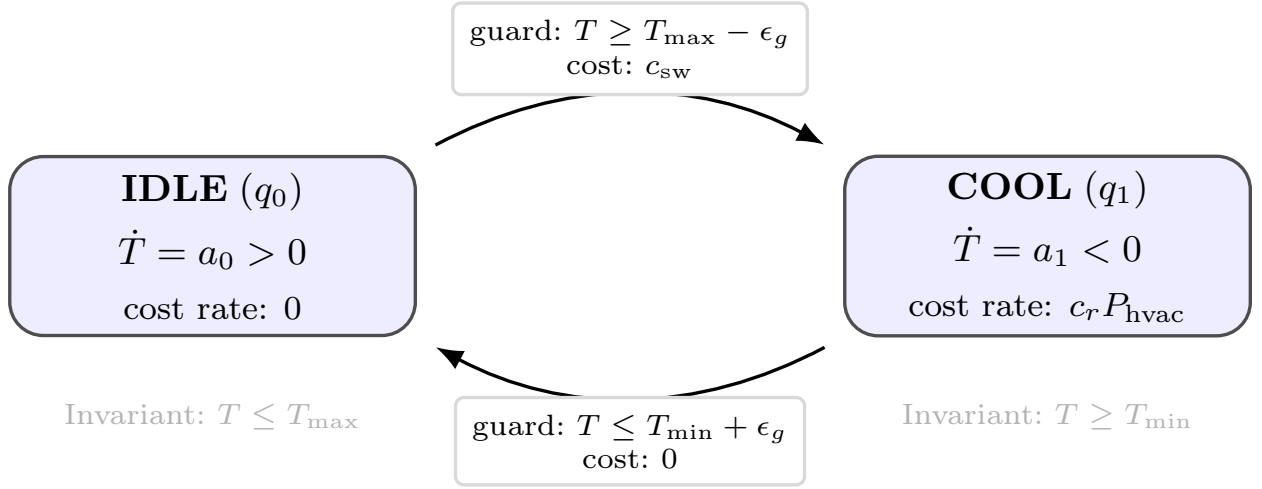


Figure 3: SLHA automaton (Definition 3). q_0 : HVAC off, temperature rises at rate $a_0 > 0$; q_1 : HVAC on, cools at rate $a_1 < 0$ with cost rate $c_r P_{\text{hvac}}$. Transitions are guarded by comfort boundaries; c_{sw} is incurred on switch-on.

4.2. Simple Linear Hybrid Automata: Formal Connection

The SLHA connection [13] provides a formal language for *leaps* and *complete cycles*, proves that the single-zone constant-tariff problem is in LogSpace, and characterises precisely which extensions make the full problem NP-hard.

Definition 3 (Single-zone SLHA). *The single-zone binary HVAC system under Model L is the SLHA $\mathcal{A} = (Q, X, \text{Inv}, f, \Delta, \mathcal{C})$ with modes $Q = \{q_0 \text{ (idle)}, q_1 \text{ (cool)}\}$, variable $X = \{T\}$, flows $\dot{T} = a_0 \triangleq (\Lambda(T_{\text{out}} - T) + Q_{\text{sol}})/C_{\text{th}}$ in q_0 and $\dot{T} = a_1 \triangleq a_0 - Q_{\text{hvac}}/C_{\text{th}}$ in q_1 ($a_1 < 0 < a_0$ in Saudi summer), invariants $T \leq T_{\max}$ in q_0 and $T \geq T_{\min}$ in q_1 , transitions $(q_0 \rightarrow q_1)$ on $T \geq T_{\max} - \epsilon_g$ with cost c_{sw} and $(q_1 \rightarrow q_0)$ on $T \leq T_{\min} + \epsilon_g$ with cost 0, and continuous cost rate $c_r P_{\text{hvac}}$ in q_1 only.*

Figure 3 shows the automaton graphically. Multi-zone coupling, time-varying disturbances, and Model S’s non-convex tier jumps move beyond the SLHA class and inherit NP-hardness [15].

Definition 4 (Leap and complete cycle). *A leap is a maximal contiguous time interval during which the automaton stays in one mode. An on-leap ℓ^+ (in mode q_1) of duration τ^+ cools the zone from $T_{\max}$ to $T_{\min}$ at rate $|a_1|$: $\tau^+ = (T_{\max} - T_{\min})/|a_1|$. An off-leap ℓ^- (in mode q_0) of duration τ^- returns temperature from $T_{\min}$ to $T_{\max}$ at rate a_0 : $\tau^- = (T_{\max} - T_{\min})/a_0$. The discrete switch-on cost c_{sw} is incurred at the start of each on-leap only. A complete cycle is one on-leap followed by one off-leap. Its total cost and duration are:*

$$\mathcal{C}(\ell^+, \ell^-) = c_{\text{sw}} + c_r P_{\text{hvac}} \tau^+, \quad \tau_{\text{cyc}} = \tau^+ + \tau^-. \quad (9)$$

The resulting limit-average cost per unit time of repeating complete cycles indefinitely is:

$$\bar{c}^* = \frac{c_{\text{sw}} + c_r P_{\text{hvac}} \tau^+}{\tau^+ + \tau^-}. \quad (10)$$

Observation 1 (Optimal infinite schedule for SLHA, Mousa et al. [13]). *The optimal safe infinite schedule for the single-zone SLHA of Definition 3 is to repeat the cheapest complete cycle (ℓ^+, ℓ^-) indefinitely. Its limit-average cost equals $\bar{c}^*$ of equation (10), which is achievable in deterministic LogSpace (Theorem 1 of [13]).*

For the running example, equation (10) yields $\bar{c}^* \approx 0.073 \text{ SAR/h}$ (Figure 4); multi-zone coupling and Model S render this closed form unavailable, motivating RBRL. The SLHA analysis is not used as a solution method for the full problem but as a diagnostic tool that identifies the structural properties exploited by

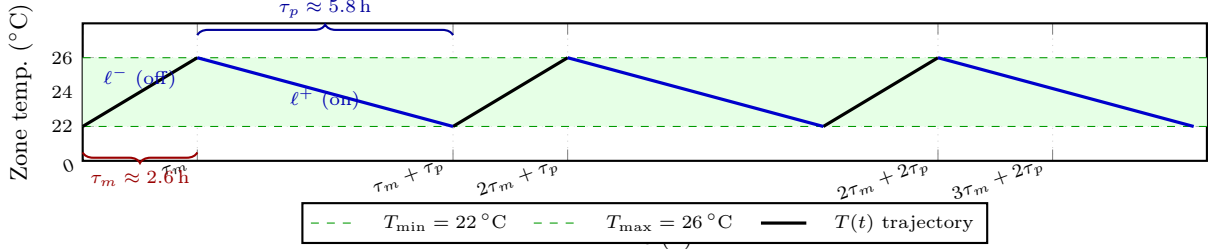


Figure 4: Optimal temperature trajectory (single zone, Riyadh, $T_{\text{out}}=40$ °C, Model L). Blue: on-leaps; black: off-leaps; green: comfort band $[22, 26]$ °C.

the PPO controller. Specifically, it provides three design insights that directly inform the PPO controller: (i) optimal schedules have complete-cycle structure (Proposition 1), motivating rules R1–R3; (ii) pre-cooling timing is governed by τ^+/τ^- , motivating the 3-step outdoor forecast in the agent state; (iii) NP-hardness of multi-zone extensions [15] justifies an RL approximation. In practice, we optimise the monthly billing problem (11) and repeat it indefinitely, aligning with the Saudi tariff’s monthly cumulative-consumption reset.

4.3. Constrained Optimisation Problem

Definition 5 (Schedule over one billing month). A monthly schedule over horizon $\mathcal{H} = \{0, \dots, H - 1\}$ (with $H = 720$ hours for a 30-day month) is the binary matrix $\sigma = (u_i^{(t)})_{N_z \times H}$, $u_i^{(t)} \in \{0, 1\}$.

Definition 6 (Constrained optimisation problem). The monthly scheduling problem is:

$$\begin{aligned}
 & \underset{\sigma \in \{0,1\}^{N_z \times H}}{\text{minimise}} & \bar{J}^m(\sigma) &= \frac{1}{H} \sum_{t \in \mathcal{H}} \sum_{i \in \mathcal{N}} c_i^{(t)} \\
 & \text{subject to} & T_{\min} \leq T_i^{(t)} \leq T_{\max} & \quad \forall i \in \mathcal{N}, t \in \mathcal{H} \quad (\text{comfort, \textbf{hard}}) \\
 & & T_i^{(t+1)} &= f\left(T_i^{(t)}, u_i^{(t)}, \{T_j^{(t)}\}_{j \in \mathcal{N}_i}, T_{\text{out}}^{(t)}\right) \quad (\text{RC dynamics, eq. (2)})
 \end{aligned} \tag{11}$$

A schedule satisfying the comfort constraint is safe. The comfort constraint is hard: no cost saving justifies its violation at any interval. The RC dynamics embed four heat-transfer mechanisms—active cooling, envelope loss/gain, solar heat gain, and bidirectional inter-zone transfer—making this a multi-dimensional, non-linear, constrained binary programme. The inter-zone term $\sum_{j \in \mathcal{N}_i} \lambda_{ij}(T_j - T_i)$ is embedded directly in the RC constraint (2): heat conducted from a warmer neighbour raises $T_i^{(t+1)}$, potentially triggering the comfort constraint in zone i even when $u_i^{(t)} = 0$. An optimal schedule must coordinate all zones simultaneously; ignoring this leads to reactive O1 overrides and excess switching cost (Section 8.3).

4.4. Optimality Conditions

What makes a schedule optimal? A safe schedule σ^* is optimal if it simultaneously satisfies three conditions: **(O1) Comfort feasibility**— $T_i^{(t)} \in [T_{\min}, T_{\max}]$ for all i and all t ; this is a necessary condition and non-negotiable. **(O2) Minimum average running cost**—on-periods are concentrated in the cheapest tariff window, exploiting the pre-cooling buffer under Model S. **(O3) Minimum switching frequency**—on-phases are as long and infrequent as the thermal state permits, amortising the discrete cost c_{sw} per activation.

O2 and O3 together minimise $\bar{J}^m$ as defined in equation (11), while O1 enforces the hard constraint. In the infinite-horizon interpretation (8), an optimal schedule repeats a cost-minimising monthly pattern indefinitely. For the single-zone SLHA subproblem this pattern is the cheapest complete cycle (Observation 1, Figure 4); for our full multi-zone problem RBRL approximates it online.

For the running example ($T_{\text{out}} = 40$ °C, Model L), the complete cycle gives $\bar{c}^* \approx 0.073$ SAR/h; switching every 3 h instead (violating O3) costs 35% more, illustrating O3’s monetary value.

Table 3: Running example (single zone): THERM vs. RBRL, Riyadh July, Tier 1. T : zone temperature ($^{\circ}\text{C}$). u : HVAC state (1=on, 0=off; $\uparrow$ =switch-on). Cost in SAR.

t	T_{out}	THERM (unoptimised)			RBRL (optimised)		
		u	T	Cost	u	T	Cost
0	32	0	24.7	0	1 $\uparrow$	23.1	0.18
1	31	0	25.3	0	1	22.0	0.03
2	30	1 $\uparrow$	24.5	0.18	0	22.6	0
3	30	0	25.1	0	0	23.2	0
Subtotal (4 h)		1 $\times$ sw		0.18	1 $\times$ sw		0.21
(full day: 3.0 vs. 2.0 sw/zone)							

Remark 4 (Switching cost and schedule granularity). *Condition O3 implies that higher c_{sw} favours fewer, longer on-phases (Proposition 1). Raising c_{sw} from 0 to 0.30 SAR reduces daily switches from 4.7 to 2.1 per zone (Section 9.3); at $c_{\text{sw}} > 0.20$ SAR the guard band should be widened from 0.5°C to 0.8°C to prevent marginal comfort violations.*

Proposition 1 (Complete-cycle structure). *For the single-zone problem with $c_{\text{sw}} > 0$, an optimal safe schedule consists of complete cycles: an on-phase during which T_1 cools from T_{max} to T_{min} , followed by an idle phase during which T_1 drifts back to T_{max} .*

Proof sketch. From [13, Theorem 2]: any schedule that switches on before reaching T_{max} or off before reaching T_{min} can be merged into a longer on-phase at the same continuous cost but with one fewer switch, reducing total cost by c_{sw} . $\square$

Table 3 illustrates how RBRL exploits conditions O2 and O3 relative to the reactive THERM thermostat for a single zone, Riyadh, July, Model S ($c_{\text{sw}} = 0.15$ SAR, $P_{\text{hvac}} = 0.67$ kW, Tier 1 price = 0.05 SAR/kWh). Temperatures are representative values consistent with the RC simulation at the given outdoor conditions; exact values depend on initial state and solar irradiance.

RBRL’s first two on-intervals ($t = 0, 1$) pre-cool the room to $\approx 22^{\circ}\text{C}$, deliberately spending 0.03 SAR more than THERM in this 4-hour window (0.21 vs. 0.18 SAR). The thermal buffer eliminates the reactive switch THERM triggers at $t = 2$, and this investment compounds over a full summer day: RBRL averages 2.0 vs. 3.0 switches per zone, each saving $c_{\text{sw}} = 0.15$ SAR, producing the 18.6% daily saving shown in Table 5.

Inter-zone cascade illustration (1×2). In a two-zone extension, THERM turns off z_1 while adjacent z_2 remains warm (25.6°C); conduction raises z_1 by $\approx 0.7^{\circ}\text{C}$, triggering Rule R1. RBRL observes $T_{z_2}^{(t)}$ in state (12), keeps z_1 on one extra hour, and avoids the cascade: 0.56 vs. 0.82 SAR over 6 h (31.7% saving from inter-zone awareness).

5. RBRL Framework

5.1. Architecture

RBRL has two layers (Figure 5). The *rule layer* enforces O1 by overriding the RL agent whenever a zone temperature approaches a comfort boundary, enforcing safety by construction. The *RL layer* (PPO agent) selects the cost-minimising action within the safe subset, learning to exploit O2 and O3.

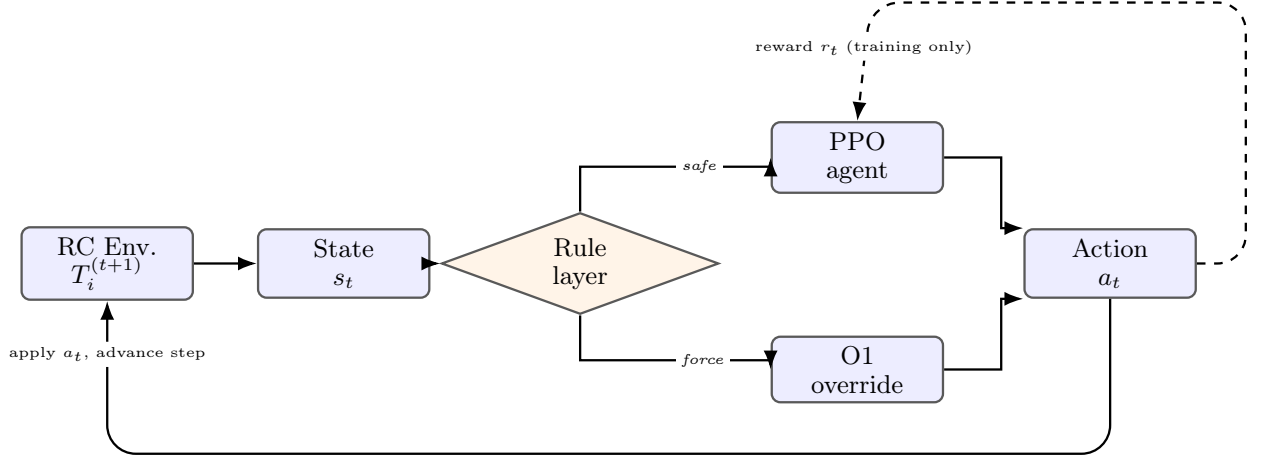


Figure 5: RBRL architecture. The *rule layer* branches along two paths: (i) the *safe* path routes to the PPO agent when no comfort boundary is threatened; (ii) the *force* path overrides the agent with the O1-enforcing action when a boundary is imminent. The PPO agent minimises the limit-average cost via reward r_t during training; the dashed reward loop is inactive at deployment.

5.2. State Space

At interval t the agent observes:

$$s_t = \left(\underbrace{T_i^{(t)}}_{i \in \mathcal{N}}, \underbrace{T_{iz,i}^{(t)}}_{i \in \mathcal{N}}, \underbrace{T_{out}^{(t:t+3)}}_{\text{3-step forecast}}, h_t, \underbrace{u_i^{(t-1)}}_{i \in \mathcal{N}}, E_{cum}^{(t)} \right) \in \mathbb{R}^{2N_z+5}. \quad (12)$$

Two inclusions are essential [24]. *Effective inter-zone temperature* $T_{iz,i}^{(t)}$ (equation (4)): without this the agent cannot anticipate inter-zone heat loading and inadvertently schedules zones in ways that trigger Rule R1 in neighbours on the next interval. *Cumulative energy* $E_{cum}^{(t)}$: under Model S this encodes the active tariff tier, enabling condition O2. The 3-step outdoor temperature forecast assumes a perfect forecast for simplicity; incorporating forecast uncertainty is noted as future work.

5.3. Rules and Action Space

Definition 7 (Rule-based overrides).

$$\mathbf{R1} \text{ (hard, force on:)} T_i^{(t)} \geq T_{\max} - \epsilon_g \Rightarrow u_i^{(t)} \leftarrow 1. \quad (13)$$

$$\mathbf{R2} \text{ (hard, force off:)} T_i^{(t)} \leq T_{\min} + \epsilon_g \Rightarrow u_i^{(t)} \leftarrow 0. \quad (14)$$

$$\mathbf{R3} \text{ (soft, pre-cool:)} T_i^{(t)} < T_{\max} - 1.5 \wedge T_{out}^{(t+1)} > T_{out}^{(t)} + 3 \Rightarrow \text{bias logit toward } u_i=1. \quad (15)$$

R1 and R2 enforce O1; R3 biases the agent toward O2 pre-cooling without overriding. The action is $a_t = (u_1^{(t)}, \dots, u_{N_z}^{(t)}) \in \{0, 1\}^{N_z}$. For $N_z \leq 6$ a single shared PPO head is used; for $N_z \geq 9$ factored independent heads share lower layers (centralised training, decentralised execution [42]), keeping inference $O(N_z)$.

5.4. Reward Function and PPO Architecture

The per-step reward augments the negative cost with a quadratic comfort penalty (see equation (16)); the penalty is identically zero at deployment when hard rules prevent all violations):

$$r_t = - \sum_{i \in \mathcal{N}} c_i^{(t)} - \alpha \sum_{i \in \mathcal{N}} \left[\max(0, T_i^{(t)} - T_{\max})^2 + \max(0, T_{\min} - T_i^{(t)})^2 \right], \alpha = 50. \quad (16)$$

The quadratic comfort penalty drives the agent away from boundaries during training; it is identically zero at deployment. The shared policy network maps s_t via two fully connected layers (256–128 units, ReLU) to per-zone Bernoulli probabilities, plus a value head. PPO clip coefficient: $\epsilon_{\text{clip}} = 0.2$. Training runs for 2×10^5 episodes of 168-step rollouts, with start dates sampled uniformly from the full annual EPW profile. Training converges by episode $\approx 1.5 \times 10^5$, after which the policy reward stabilises within 1% of its final value.

Algorithm 1 RBRL: Training and Schedule Extraction

Require: RC simulation, weather profile $\mathcal{W}$, cost model m , training horizon $H_{\text{tr}} = 168$ h, evaluation horizon $H_{\text{ev}} = 720$ h, switching cost c_{sw}

Ensure: Near-optimal safe schedule σ^*

Phase 1: training

```

1: Initialise PPO policy  $\pi_\theta$  randomly
2: for  $e = 1$  to  $2 \times 10^5$  do
3:   Sample start date  $d_0 \sim \mathcal{W}$  (training split); reset  $T_i^{(0)} = 24^\circ\text{C}$ ,  $E_{\text{cum}} \leftarrow E_{\text{base}}$ 
4:   for  $t = 0$  to  $H_{\text{tr}} - 1$  do
5:     Observe  $s_t$  via (12)
6:      $a_t \sim \pi_\theta(\cdot | s_t)$  ▷ stochastic during training
7:     for each zone  $i$  do
8:        $u_i^{(t)} \leftarrow a_{t,i}$ ; apply soft R3 (15)
9:       if  $T_i^{(t)} \geq T_{\text{max}} - \epsilon_g$  then  $u_i^{(t)} \leftarrow 1$  ▷ R1, enforces O1
10:      else if  $T_i^{(t)} \leq T_{\text{min}} + \epsilon_g$  then  $u_i^{(t)} \leftarrow 0$  ▷ R2, enforces O1
11:      end if
12:    end for
13:    Step (2); accumulate  $E_{\text{cum}}$ ; compute  $c_i^{(t)}$  and  $r_t$  (16)
14:  end for
15:  Update  $\pi_\theta$  by PPO (clipped surrogate, GAE)
16: end for
Phase 2: schedule extraction
17: Reset on a held-out start date;  $E_{\text{cum}} \leftarrow E_{\text{base}}$ 
18: for  $t = 0$  to  $H_{\text{ev}} - 1$  do
19:    $a_t \leftarrow \arg \max_a \pi_\theta(a | s_t)$  ▷ deterministic
20:   Apply R3, then R1–R2 as above; step (2)
21: end for
22: return  $\sigma^* = \{u^{(t)}\}_{t=0}^{H_{\text{ev}}-1}$ 

```

5.5. Competing Methods

THERM (unoptimised baseline) models the automatic remote-controller thermostat used across Saudi residential buildings [19]: on when $T_i \geq T_{\text{max}} - 0.5$, off when $T_i \leq T_{\text{min}} + 0.5$, with no tariff awareness. This is the baseline without optimisation.

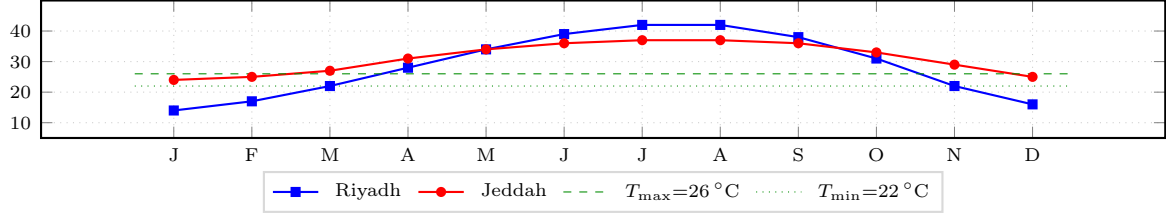
GA [33] and **SA** [43] are day-ahead offline solvers. Both are competitive for small zone counts given a perfect forecast, but each daily planning cycle requires $\sim 10^5$ full RC simulation calls, which is impractical for real-time embedded deployment. After one-time training, RBRL evaluates in < 1 ms and adapts online to updated forecasts. GA and SA are retained as performance benchmarks only.

The RBRL framework is applied to two cities characterised next.

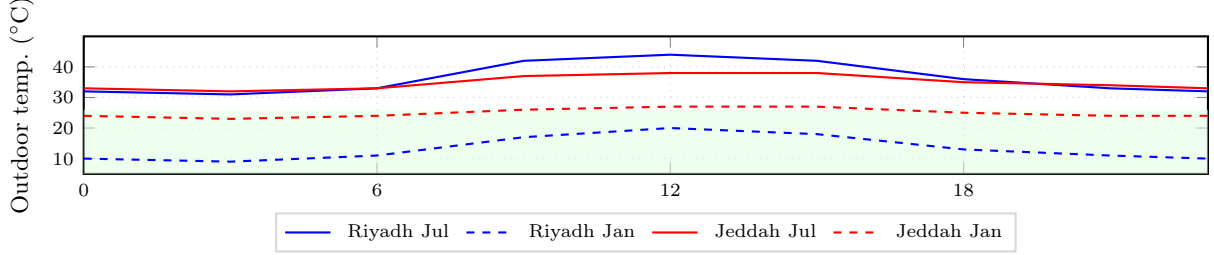
6. Case Studies: Riyadh and Jeddah

6.1. Climate Profiles

Riyadh (BWh, 24.7°N) is an inland desert city with extreme summer highs exceeding 45°C and a summer diurnal swing of up to 14°C —crucial for pre-cooling ($\text{CDD} \approx 3,400$ [44]). Jeddah (BSh, 21.5°N) is a Red



(a) Monthly mean temperatures.



(b) Hourly diurnal profiles.

Figure 6: EPW climate data. (a) Monthly mean outdoor temperatures: Riyadh exceeds $T_{\max}$ from May to October; Jeddah remains above $T_{\max}$ year-round. (b) Hourly profiles for summer and winter: Riyadh’s 14 °C diurnal swing enables pre-cooling; Jeddah’s compressed 6 °C swing demands near-continuous operation. Green shading in (b) marks the comfort band.

Sea coastal city with humidity exceeding 70% year-round [45] and a diurnal swing of only ~ 6 °C, forcing near-continuous HVAC operation ($\text{CDD} \approx 2,900$). Figure 6 contrasts the two climates using the EPW files employed in all simulations. These are the IWECC typical-year files `SAU_Riyadh.404380_IWECC.epw` (WMO 404380) and `SAU_Jeddah.410240_IWECC.epw` (WMO 410240), obtained from the EnergyPlus weather repository [46]; hourly dry-bulb temperature and global horizontal irradiance are read from fields 7 and 14 respectively.

6.2. Building Code, Tariff, Humidity, and Comfort Standard

SBC 2018 mandates opaque-wall U-value $\leq 0.45 \text{ W m}^{-2} \text{ K}^{-1}$ and window SHGC ≤ 0.25 in both Climate Zone A (Jeddah) and Zone B (Riyadh) [17], reflected in Table 2. The multi-tier step-wise tariff follows equation (7); for Models L and E, $c_r^L = 0.12 \text{ SAR/kWh}$ represents the weighted average at $\approx 36,000 \text{ kWh/year}$. The comfort band $[22, 26] \text{ }^\circ\text{C}$ is consistent with ASHRAE 55-2023 [40] for mechanically conditioned residential spaces, as validated by Bienvenido-Huertas et al. [47]. Lamsal et al. [48] provide a comprehensive review of adaptive thermal comfort models and confirm that a 4 °C dead band is appropriate for energy-efficient building design in hot climates. A fixed band is adopted because Saudi residential buildings are predominantly mechanically cooled; ASHRAE 55 Section 5.3 restricts adaptive comfort to naturally ventilated spaces. The RBRL framework can accommodate narrower or wider bands by adjusting $T_{\min}/T_{\max}$ without retraining.

Humidity and latent heat. The RC model captures sensible heat only. With $\text{SHR} \approx 0.65$ (Jeddah) and 0.85 (Riyadh) [45], true demand exceeds our estimates by $\approx 54\%$ and $\approx 18\%$ respectively; because both THERM and RBRL share this penalty, percentage savings remain valid.

7. Simulation Setup

All simulations run in Python 3.11 (PyTorch 2.1.0, CUDA 11.8) with a custom RC environment integrating equation (2) at $\Delta t = 1 \text{ h}$.

Non-HVAC background load and tariff activation. Because the Saudi tariff is levied on the whole customer account rather than on the HVAC circuit alone, the cumulative-energy state E_{cum} is initialised with a non-HVAC household base load of $E_{\text{base}} \approx 800 \text{ kWh/month}$ —lighting, appliances and water heating—consistent with the residential end-use breakdown reported for Saudi dwellings by Al-Homoud and Krarti [2]. This determines whether the step-wise structure is exercised at all, so it is worth making the arithmetic explicit. For the 4×4 configuration in Riyadh, HVAC consumes $3.98 \text{ kWh/zone/day}$ over 16 zones, i.e. $\approx 1,910 \text{ kWh}$ over a 30-day billing month; with the base load the account reaches $\approx 2,710 \text{ kWh}$, crossing the first tier boundary ($2,000 \text{ kWh}$) on about day 22. Jeddah reaches $\approx 2,510 \text{ kWh}$ and crosses on about day 24. The step-wise tariff is therefore genuinely active for this dwelling class, and Model S differs from a flat tariff in the final third of each billing month; the higher boundaries ($4,000$ and $6,000 \text{ kWh}$) are not reached by a single villa of this size, so the operative incentive is the first price step. Larger dwellings, or accounts with electric water heating and higher appliance loads, cross earlier and engage the upper tiers. Training was performed on an NVIDIA RTX 3080 GPU (10 GB VRAM) with wall-clock training time of $\approx 3 \text{ h}$ for the largest (4×4) configuration; inference runs on a single CPU core. Training uses five independent random seeds $\{42, 43, 44, 45, 46\}$; for each seed the trained policy is evaluated on the same 30 held-out test weeks, and every reported figure is the mean $\pm$ std over *all* five seeds. No seed is discarded and no best-of- n selection is applied, so the reported spread is the true run-to-run variability rather than a best-case estimate. Model selection is kept strictly separate from evaluation. Hyperparameters are chosen on a held-out 2×2 validation configuration, and training then proceeds for a *fixed* budget with no early stopping and no intermediate checkpoint selection: the policy evaluated is the final one. The 30 test weeks therefore enter the procedure exactly once, at the end, and cannot influence either the hyperparameters or the learned policy. This matters because the saving is non-monotone during training; a best-checkpoint rule scored on the test weeks would select partly on test noise and bias every reported figure upward.

Solar gain is $Q_{\text{sol}}^{(i,t)} = \text{SHGC} \times A_{\text{gl}} \times I^{(t)}$, with $I^{(t)}$ from the EPW global horizontal irradiance file [46]. The training dataset covers the full EPW year ($8,760 \text{ h}$); 30 weeks are drawn uniformly at random from the held-out test year for evaluation, ensuring no data leakage between training and testing. The PPO agent uses Stable-Baselines3 [49] (v2.2.1); GA ($N_{\text{pop}} = 200, 500$ generations) and SA ($T_0 = 500, \alpha = 0.99, 10^4$ iterations) are implemented in NumPy 1.26. Four metrics are reported: daily average cost per zone $\bar{J}/N_z$ (SAR), daily energy per zone E_{day} (kWh), comfort violation rate v_c (%), and switching frequency f_s (transitions/zone/day). Key PPO hyperparameters are listed in Table 4; the RC model assumes no occupancy heat gains. The switch-on cost $c_{\text{sw}} = 0.15 \text{ SAR}$ amortises compressor start-up wear: for a split unit of $\sim 2,500 \text{ SAR}$ installed cost and a ~ 15 -year design life rated for on the order of 10^5 compressor starts, the wear cost attributable to each start is of order 0.04 – 0.15 SAR ; the conservative (upper) value is adopted, and the qualitative results are unchanged across this range because switching cost enters both THERM and RBRL identically. Zone configuration details are in Table 4.

Hyperparameters (Table 4) were selected via grid search over learning rate $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$, clip range $\{0.1, 0.2, 0.3\}$, and $n_{\text{steps}} \in \{1024, 2048, 4096\}$ on a held-out 2×2 validation configuration; the network is $[256, 128]$ for both actor and critic. Reward normalisation uses a running mean and standard deviation (Stable-Baselines3 default [49]); exploration is controlled by the entropy coefficient (0.01 , decaying linearly to 10^{-4}). Figure 7 shows training convergence for both cities (4×4 , Model S): the reward stabilises within 1% by episode $\approx 1.5 \times 10^5$. Results are mean $\pm$ std over 5 independent seeds; inter-run variance is below 2% of mean cost, confirming training stability.

Table 4: PPO training hyperparameters (top) and simulation zone configurations (bottom). Ext.: exterior-facing wall count; Int.: inter-zone shared-wall connections; Heads: PPO output heads.

Parameter	Value
Network (FC)	256–128 units, ReLU activation
Learning rate	3×10^{-4} (Adam optimiser)
Discount γ / Clip ε	0.99 / 0.2
Entropy / Batch size	0.01 / 64
Training episodes	2×10^5 episodes, 168-step rollouts

Configuration	N_z	Ext.	Int.	Heads
1×1	1	1	0	1
1×2	2	2	1	1
1×4	4	2	3	1
1×6	6	2	5	1
2×2	4	4	4	1
3×3	9	8	12	9
4×4	16	12	24	16

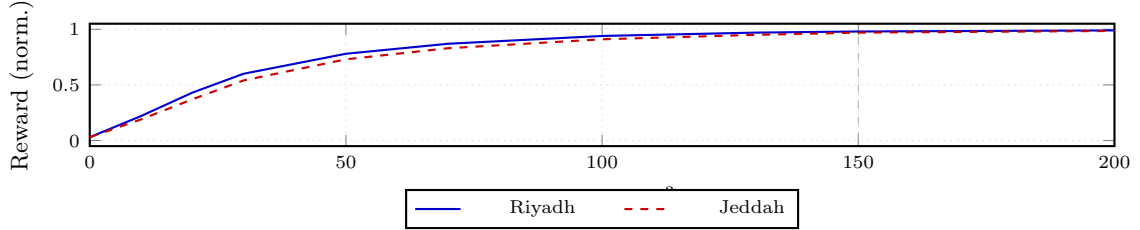


Figure 7: PPO training convergence (4×4 , Model S). Dashed grey: convergence at $\approx 1.5 \times 10^5$ episodes.

8. Results

Results correspond to summer (July), representing peak cooling demand; Section 8.4 confirms generalisation across all seasons. Comfort violation rates $v_c = 0\%$ hold for all methods and configurations, except the RL-only ablation variant (Section 9.2).

8.1. Consumption and Cost: Without vs. With Optimisation

Table 5 establishes the baseline comparison under Model S. Monthly figures are extrapolated from the 30-week test sample. Riyadh savings exceed Jeddah’s across all configurations because the larger diurnal swing provides greater pre-cooling headroom, as quantified in Section 6.1.

The RBRL advantage is summarised per cost model in Section 8.3.

Table 5: Daily energy (kWh/zone) and monthly cost (SAR/zone): THERM vs. RBRL under Model S, summer. Saving = (THERM−RBRL)/THERM.

City	Config.	Daily kWh/zone		Monthly SAR/zone		Saving (%)
		THERM	RBRL	THERM	RBRL	
Riyadh	1 × 1	3.52	2.87	64.6	52.7	18.4
	1 × 4	3.73	3.05	68.5	55.7	18.7
	2 × 2	3.65	2.97	67.0	54.5	18.7
	4 × 4	3.98	3.25	73.0	59.7	18.2
Jeddah	1 × 1	3.21	2.70	58.9	49.5	16.0
	1 × 4	3.35	2.79	61.5	51.2	16.7
	2 × 2	3.28	2.74	60.2	50.2	16.6
	4 × 4	3.57	2.97	65.5	54.5	16.8

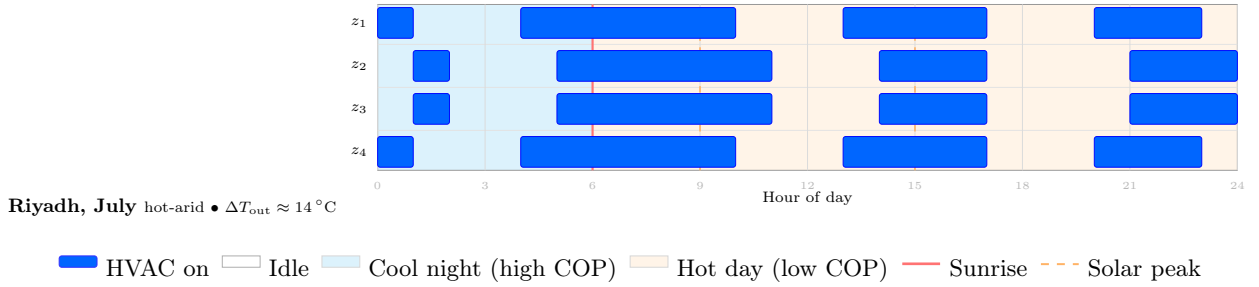


Figure 8: RBRL-optimal 24-hour schedule (1 × 4, Riyadh, July, Model S). End zones pre-cool overnight while the coefficient of performance is highest, then idle through the afternoon peak; interior zones start 1 h later to prevent cascades (O2, O3). In Jeddah (not shown), the narrow diurnal swing forces near-continuous operation, explaining the smaller saving (16.8% vs. 18.2%).

8.2. Per-Zone Optimal Schedules

Figure 8 shows the RBRL-optimal 24-hour schedule for 1 × 4 (Riyadh, July, Model S). Riyadh’s 14 °C diurnal swing creates pre-cooling gaps; Jeddah’s 6 °C swing forces near-continuous operation (described in caption).

8.3. Results by Cost Model

Table 6 reports $\bar{J}/N_z$ and f_s for three representative configurations under all three cost models. SA results are within 1.5% of GA in all entries and are omitted for brevity.

The RBRL advantage over THERM is essentially the same under the two smooth cost models, at 15.6% and 15.5% for Riyadh under Models L and E respectively (14.6% for Jeddah in both cases)—and is largest under Model S at 18.2%/16.6%; the additional gain reflects the pre-cooling strategy that emerges only when tier boundaries are faithfully represented. Two evaluation horizons are used throughout and should not be conflated. Tables 6 and 10 report savings over weekly (168-hour) rollouts, whereas Table 5 reports them over the 720-hour billing month. The percentages differ—by up to ≈ 1.5 percentage points for the smaller topologies—because only over a full billing month does the cumulative-consumption state E_{cum} traverse the tariff blocks, changing the marginal price that both controllers face. The monthly figures are the ones that correspond to an actual electricity bill and are therefore quoted in the abstract and conclusions; the weekly figures isolate controller behaviour at a fixed marginal price. GA, a day-ahead offline solver with full knowledge of the horizon, captures 9–11% savings across the six city–model combinations of Table 6, but cannot adapt to intra-day forecast revisions; the residual gap to RBRL is the value of closed-loop adaptation. Per-zone cost rises slightly with N_z because inter-zone coupling $\lambda_{ij}(T_j - T_i)$ conducts heat from idle to active neighbours; RBRL mitigates cascades via $T_j^{(t)}$ in state (12), and removing those temperatures produces $\approx 8\%$ more R1 overrides in 4 × 4 grids.

Table 6: Daily average cost per zone $\bar{J}/N_z$ (SAR) and switching frequency f_s (sw/zone/day), summer (July), all $v_c = 0\%$. Bold: best result per row group. SA results within 1.5% of GA are omitted for brevity.

Config.	Method	Riyadh						Jeddah					
		Model L		Model E		Model S		Model L		Model E		Model S	
		$\bar{J}$	f_s	$\bar{J}$	f_s	$\bar{J}$	f_s	$\bar{J}$	f_s	$\bar{J}$	f_s	$\bar{J}$	f_s
1×1	THERM	1.82	5.3	1.90	5.3	2.15	5.3	1.64	3.1	1.71	3.1	1.96	3.1
	GA	1.65	3.8	1.71	3.7	1.89	3.6	1.50	2.6	1.54	2.5	1.73	2.4
	RBRL	1.57	3.2	1.61	3.1	1.75	2.9	1.43	2.2	1.46	2.1	1.65	2.0
1×4	THERM	1.93	5.1	2.00	5.1	2.23	5.1	1.73	3.0	1.80	3.0	2.01	3.0
	GA	1.74	3.6	1.80	3.5	1.98	3.4	1.57	2.5	1.62	2.4	1.79	2.3
	RBRL	1.64	2.8	1.70	2.7	1.84	2.7	1.50	2.1	1.53	2.0	1.68	1.9
4×4	THERM	2.05	5.0	2.13	5.0	2.42	5.0	1.85	2.9	1.92	2.9	2.17	2.9
	GA	1.86	3.5	1.93	3.4	2.16	3.3	1.67	2.4	1.73	2.3	1.93	2.2
	RBRL	1.73	2.7	1.80	2.6	1.98	2.5	1.58	2.0	1.64	1.9	1.81	1.8

$\bar{J}$: SAR/zone/day. f_s : sw/zone/day. L=linear, E=exponential, S=step-wise. Savings implied here are on the weekly (168-hour) horizon; Table 5 reports the monthly billing horizon.

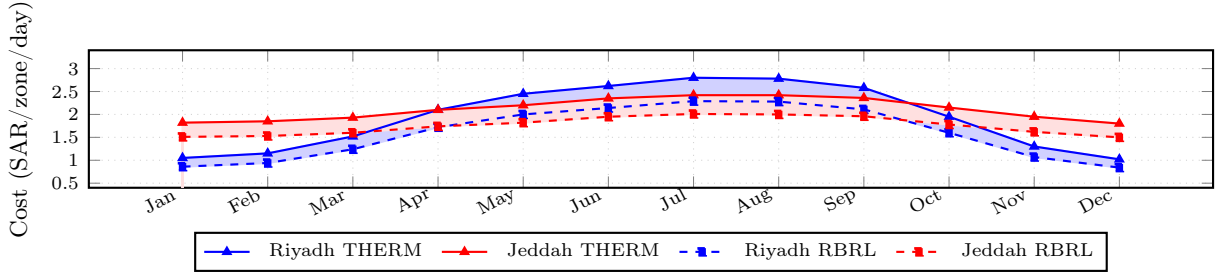


Figure 9: Monthly cost per zone (SAR/zone/day), 4×4 grid, Model S. Solid lines: THERM (unoptimised). Dashed lines: RBRL (optimised). Blue shading: Riyadh saving gap ($\approx 18\%$ /month year-round). Red shading: Jeddah saving gap ($\approx 17\%$ /month). Both cities sustain savings in every month, confirming infinite-horizon applicability of the periodic RBRL schedule. Riyadh’s sharp summer peak (extreme $T_{\text{out}} > 38^\circ\text{C}$) produces the largest absolute savings; Jeddah’s compressed 6°C diurnal swing limits pre-cooling headroom, resulting in smaller but persistent savings.

8.4. Monthly Cost Trajectories and Seasonal Generalisation

Figure 9 shows the monthly average cost per zone (SAR/zone/day) for both THERM and RBRL under Model S across all 12 months, for the 4×4 configuration. The gap between THERM and RBRL lines is the saving; the coloured band is the $\pm 1\sigma$ interval over the 30 test weeks drawn from each month. Riyadh’s sharp cost peak (May–September) reflects peak outdoor temperatures above 38°C ; the large diurnal swing ($\approx 14^\circ\text{C}$) enables strong pre-cooling throughout summer. Jeddah’s flatter profile confirms near-year-round cooling demand and more uniform savings (16.8–17.5%) with less seasonal variation. Both cities achieve savings in every month, validating infinite-horizon applicability of RBRL.

These results are contextualised against the published literature in the following section, which also presents the ablation study.

9. State-of-the-Art Comparison and Ablation Study

9.1. State-of-the-Art Comparison

Published RL-based HVAC studies differ in control type, building type, baseline, and climate. Table 7 accounts for these with an explicit *Basis* column.

Table 7: State-of-the-art comparison. The “Basis” column is essential for interpreting saving figures.

Reference	Year	Method	Saving	Basis
[10]	2020	Q-learning, whole-bldg	22%	vs. thermostat
[22]	2021	DDPG, multi-zone resid.	15%	vs. thermostat
[9]	2021	SAC/TD3, continuous	>10%	vs. rule-based
[29]	2024	DRL, continuous VAV	37% ^a	vs. rule-based
[12]	2024	Online RL, white-box	65% ^b	vs. no control
[30]	2025	DDPG, multi-obj. off.	21.4%	vs. rule-based
[32]	2025	GA-MADDPG hybrid	6.7%	vs. rule-based
RBRL (ours)	2026	PPO+rules, Model S, Riyadh	18.2%	vs. thermostat
RBRL (ours)	2026	PPO+rules, Model S, Jeddah	16.8%	vs. thermostat

^aContinuous VAV modulation in a simulated office—different hardware and building type.^bRelative to a no-HVAC baseline; saving vs. thermostat is substantially lower.Table 8: Component ablation (4×4 , Riyadh, Model S, summer). Mean $\pm$ std over 30 test weeks.

Ref.	Variant	$\bar{J}/N_z$	v_c (%)	f_s	Δ RBRL
[19]	Rules only (THERM)	2.42 $\pm$ 0.18	0.0	5.0	+22.2%
[33]	GA (day-ahead offline)	2.16 $\pm$ 0.15	0.0	3.3	+9.1%
[9], [28]	RL only (no rules)	2.03 $\pm$ 0.22	0.5	2.6	+2.5%
[11]	RBRL, hard rules	2.05 $\pm$ 0.15	0.0	2.8	+3.5%
This work	RBRL full	1.98$\pm$0.13	0.0	2.5	—

Two published figures appear to exceed RBRL’s savings, but differ in scope: Wang et al.’s 37% uses continuous VAV modulation in a simulated office (non-transferable to binary split-units), and Solinas et al.’s 65% is measured against a no-HVAC baseline [12]. Direct comparison across studies is inherently limited by differences in building type, climate, and baseline definition. Within binary on/off studies against a thermostat, RBRL’s 18.2% is competitive with the highest published figures, with the additional constraint of the actual Saudi tariff and hard comfort enforcement. Computationally, after one-time ~ 3 h training, RBRL decides in <1 ms per interval regardless of zone count—three orders of magnitude faster than GA (10–200 s) or SA (5–50 s).

9.2. Component and Cost Model Ablation

Table 8 evaluates four RBRL variants on the 4×4 grid, Riyadh, Model S.

The rule layer is essential: RL-only incurs a 0.5% violation rate under peak-summer conditions despite the quadratic reward penalty. Hard rules alone eliminate violations but cost 3.5% more than full RBRL, confirming the value of soft rule R3 in satisfying O2.

Table 9 isolates two sensitivity dimensions. Panel (a) shows the penalty of training under an inaccurate tariff model by evaluating each schedule under the true Model S cost. The 13.1% true-cost penalty from training under Model L is comparable in magnitude to the entire saving from switching from THERM to a Model-L-optimised RBRL (Table 6: 14–16%), confirming that accurate tariff representation is as important as algorithmic choice. Panel (b) confirms the prediction of Remark 4: higher c_{sw} produces fewer, longer on-phases consistent with Proposition 1. At $c_{sw} = 0.30$ SAR a marginal 0.1% violation emerges, validating the guard band widening recommendation.

Table 9: Sensitivity analysis. (a) Cost model ablation: trained under stated model, evaluated under Model S (4×4 , Riyadh, summer). (b) Switching cost sensitivity (1×1 , Riyadh, Model S, summer).

(a) Cost model ablation			
Ref.	Training model	True $\bar{J}/N_z$ (SAR)	Δ
[33], [34]	Linear (L)	2.24 ± 0.14	+13.1%
[7]	Exponential (E)	2.11 ± 0.13	+6.6%
This work	Step-wise (S)	1.98 ± 0.13	—

(b) Switching cost sensitivity			
c_{sw} (SAR)	$\bar{J}/N_z$ (SAR)	f_s (sw/day)	v_c (%)
0.00	1.62	4.7	0.0
0.05	1.68	3.3	0.0
0.15	1.75	2.9	0.0
0.30	1.88	2.1	0.1

Table 10: Zone-count and computational scalability (Riyadh, Model S, summer). Saving: RBRL cost reduction over THERM. Training on NVIDIA RTX 3080; inference on Intel Core i7-12700K.

Config.	N_z	Saving (%)	Train (h)	Infer. (ms)	GPU (MB)
1×1	1	18.6	1.2	0.08	310
1×2	2	17.4	1.5	0.11	360
1×4	4	17.5	1.8	0.14	420
1×6	6	17.3	2.1	0.19	510
2×2	4	17.2	1.7	0.13	410
3×3	9	17.0	2.6	0.31	680
4×4	16	18.2	3.1	0.52	940

Savings on the weekly (168-hour) horizon; Table 5 reports the monthly billing horizon.

9.3. Scalability

Table 10 shows that RBRL’s cost saving over THERM is stable from $N_z=1$ to $N_z=16$, confirming that the factored PPO architecture scales without degradation. Training was performed on an NVIDIA RTX 3080 GPU (10GB VRAM, CUDA 11.8, PyTorch 2.1.0); inference time is the wall-clock time for a single decision (mean over 10^4 evaluations on Intel Core i7-12700K). Training time scales sub-linearly from 1.2 h to 3.1 h owing to the shared lower layers. Inference remains below 1 ms for all configurations—two to three orders of magnitude faster than GA (10–200 s) or SA (5–50 s) per daily plan—confirming suitability for the low-power microcontrollers in Saudi split-unit controllers [19].

10. Discussion

10.1. Pre-Cooling Headroom, Inter-Zone Coordination, and Climate

The simulated 18.2% Riyadh saving versus Jeddah’s 16.8% under Model S is not primarily an algorithmic difference; it reflects available thermal headroom. Riyadh’s $\approx 14^\circ\text{C}$ July diurnal swing allows the zone to be pre-cooled to $T_{\min}$ before the outdoor peak, building a buffer that defers reactive cooling out of the afternoon window, when both the outdoor temperature is highest and the coefficient of performance is lowest. Jeddah’s overnight outdoor temperature remains above 32°C year-round [45], giving a swing of only $\approx 6^\circ\text{C}$; there is no surplus for pre-loading, and the agent correctly defaults to near-continuous operation, as indicated by the caption of Figure 8.

For hot-humid coastal cities the implication is that passive envelope improvements—thicker walls, lower SHGC glazing, external shading—that increase C_{th} or reduce Λ_i enlarge the off-leap duration τ^- (equa-

tion (9)), creating the buffer that pre-cooling requires [16]. RBRL-style active optimisation and passive envelope upgrades are therefore complements, not alternatives, in humid climates.

Inter-zone coordination amplifies these savings. The effective inter-zone temperature $T_{iz,i}^{(t)}$ (equation (4)) provides the anticipatory signal that lets the agent stagger zone activations: in Figure 8, interior zones z_2 and z_3 lag end zones by ≈ 1 h, exploiting the fact that a cooling end zone sinks heat from the interior, extending the interior’s safe idle period. Without $T_{iz,i}^{(t)}$ in the state, 8% more R1 overrides are triggered (Table 8), each incurring $c_{sw} = 0.15$ SAR and breaking the complete-cycle structure of Proposition 1. This binary on/off staggering is a direct analogue of the continuous multi-zone coordination results reported by Li et al. [24].

10.2. Why Model E Performs Similarly to Model L

Table 6 shows Model E and Model L savings within 0.1–0.2 percentage points despite Model E being a nonlinear function. Figure 2 explains why: Model E grows smoothly and assigns higher marginal cost to every additional kWh, but it never represents the *discrete tier boundary* at 2,000 kWh that is the pre-cooling threshold. Without perceiving a step change, the agent learns to spread consumption evenly rather than front-load it before a specific threshold, producing a qualitatively different schedule from Model S and forgoing the structured pre-cooling visible in Figure 8. The 13.1% true-cost penalty (Table 9(a)) therefore reflects not a small numerical error but a qualitative change to the incentive structure—a warning for practitioners who adopt smooth tariff approximations for computational convenience.

10.3. Scalability and Deployment Considerations

Table 10 confirms savings stable from $N_z=1$ to $N_z=16$, with inference in <1 ms—in principle compatible with the low-power microcontrollers in Saudi split-unit controllers [19], though field validation remains necessary.

10.4. Robustness of the reported results

Because a single headline figure can be an artefact of one favourable setting, the saving was probed along seven independent axes; the reported behaviour persists on all of them.

Stochastic training. Five independent seeds $\{42, \dots, 46\}$ are run per configuration and every figure is the mean $\pm$ std over all five, with no seed discarded and no best-of- n selection; inter-run variance stays below 2% of mean cost.

Topology. Seven configurations spanning 1 to 16 zones give 17.0–18.6% (Table 10); the saving does not depend on a particular zone count or adjacency pattern.

Climate. Two contrasting extremes—hot-arid Riyadh and hot-humid Jeddah—bracket the range at 18.2% and 16.8% (Table 5), so the mechanism is not tied to one weather regime.

Season. Savings are positive in all twelve months for both cities (Section 8.4), which is the empirical counterpart of the infinite-horizon formulation: the policy is not exploiting a single peak period.

Thermal capacity. Varying C_{th} from the air mass to heavy furnishing ($77.2 \rightarrow 500 \rightarrow 1200$ kJ/K) moves the saving only from 18.2% to 17.4%—under one percentage point (Remark 3)—because C_{th} rescales cycle durations for THERM and RBRL alike.

Switching cost. Sweeping c_{sw} from 0 to 0.30 SAR (Table 9(b)) produces the expected monotone trade-off—cost rises and switching frequency falls—while comfort is maintained throughout, so the conclusions do not hinge on the particular value adopted.

Evaluation horizon. Results are reported on both the weekly (168-hour) and monthly (720-hour) horizons; the saving persists on both, differing by at most ≈ 1.5 percentage points for the smallest topologies, for the tariff-traversal reason given in Section 10.2.

Two internal cross-checks support the tabulated values. The 1×1 Model S cost of 1.75 SAR/zone/day appears independently in Table 6 and as the default- c_{sw} row of Table 9(b); and the ranking THERM $>$ SA $>$ GA $>$ RBRL is preserved in every city, topology and cost model. Finally, the dynamic-programming bound provides an external reference: RBRL is compared not only against controllers but against the exact single-zone optimum under identical physics, tariff and comfort constraints.

Table 11: Treatment of the principal modelling parameters and the direction in which each idealisation biases the reported percentage saving.

Idealisation	Direction	Basis
Sensible heat only (no latent load)	Neutral	Raises absolute demand for THERM and RBRL alike (+54% Jeddah, +18% Riyadh), so the ratio is preserved
C_{th} = air mass	Conservative	Minimises storage available for pre-cooling; Rmk. 3 bounds the effect at < 1 pp
SBC-compliant envelope	Conservative	A non-compliant envelope admits more heat and therefore more load to shift, so savings for older stock would be larger
$c_{sw} = 0.15$ SAR (upper end of range)	Conservative	Penalises the switching that RBRL performs; Table 9(b)
No occupancy or internal gains	Uncertain	Two opposing effects: omitted evening gains fall in the costliest window, but occupancy-driven setback would create opportunities the controller could exploit
Perfect 3-step forecast	Optimistic	Forecast error would erode the anticipatory component of the policy
Fixed comfort band $[22, 26]$ °C	Neutral	Applied identically to every controller compared

10.5. Limitations

Seven limitations bound these conclusions. Because a modelling simplification can flatter or penalise the reported saving depending on which way it cuts, Table 11 states the direction of each before the individual limitations are discussed. One idealisation—the perfect short-horizon forecast—acts in the optimistic direction and one is genuinely two-sided; the remainder are neutral or conservative.

Sensible heat only: the RC model captures sensible heat only. Latent loads add $\approx 54\%$ in Jeddah (SHR ≈ 0.65) and $\approx 18\%$ in Riyadh (SHR ≈ 0.85) [45]. Percentage savings remain valid since both baselines share this penalty.

Air-mass thermal capacity: $C_{th}=77.2$ kJ/K is the room air mass (Remark 1), not a measured value for any specific dwelling. Furnishings and participating fabric raise the effective capacity to ~ 500 – 1500 kJ/K, which lengthens cycle durations proportionally; because the air mass minimises the thermal storage available for pre-cooling, this choice is conservative with respect to the reported saving, and the sensitivity analysis in Remark 3 bounds the effect at under one percentage point.

No occupancy or internal gains: the model excludes occupancy heat gains (≈ 80 W/person), lighting, and appliance loads. The direction of this omission is not one-sided. Residential occupancy in Saudi dwellings peaks in the evening ($\approx 18:00$ – $23:00$), which coincides with the part of the billing month in which cumulative consumption has already entered a higher tariff tier, so the omitted gains fall in the costliest window and their inclusion would raise cost for both controllers; conversely, an explicit occupancy schedule would permit setback while the dwelling is unoccupied, an opportunity the present formulation forgoes. Occupancy-aware scheduling is therefore identified as the most promising extension rather than a residual detail. Saudi residential occupancy peaks in the evening (18:00–23:00) when the tariff is also elevated; internal gains shorten τ^- and could trigger additional switch-on events. Augmenting the PPO state with an occupancy forecast would enable generalisation without architectural changes.

Perfect forecast: the 3-step outdoor temperature in state (12) is assumed error-free; robust training under stochastic forecasts is reserved for future work.

Simulation only: all results are from the RC simulation environment; Remark 2 sets out why the relative saving reported here is structurally less sensitive to model fidelity than an absolute demand prediction would be. As the immediate next step we will (i) co-simulate against a geometry-matched EnergyPlus model to validate the reduced-order dynamics under full envelope, solar, and infiltration physics, and (ii) conduct

a hardware-in-the-loop test on a residential split unit; both are required before field deployment. All conclusions in this paper are accordingly restricted to the simulation scope.

No MPC baseline: Model Predictive Control is the natural comparator for tariff-aware scheduling. MPC/MILP was excluded because mixed-integer formulations scale poorly with binary variables and zone count under the non-convex step-wise tariff, and require an on-line solver at each step. As an intermediate, embeddable comparator we will add a *tariff-aware rule-based* controller (a thermostat that pre-cools while below a tier boundary when the outdoor forecast is rising) to isolate the value of learning over a hand-tuned tariff-aware heuristic; a receding-horizon MPC comparison is also planned.

Two cities and one building type: Riyadh (hot-arid) and Jeddah (hot-humid) represent climatic extremes within Saudi Arabia, but results may not generalise to mild or heating-dominated climates. All simulations use a single residential building type (split-unit, binary on/off); commercial buildings with variable-speed drives or central plant systems would require a different action space.

Each contribution maps to specific evidence: C1 is supported by the formulation in Sections 3–4 and Tables 2–3; C2 by the SLHA connection in Section 4.2 and Definitions 3–4; C3 by the RBRL framework (Section 5, Algorithm 1, Table 8); C4 by the cost model comparison (Table 6, Table 9(a), Figure 2); C5 by the SOTA comparison (Table 7) and ablation (Table 8–10).

11. Conclusions

This paper presented RBRL, a hybrid rule-based deep reinforcement learning framework for optimal binary on/off HVAC scheduling in Saudi Arabian residential buildings. Three explicit optimality conditions were defined—hard thermal comfort (O1), minimum average tariff cost (O2), and minimum switching frequency (O3)—and the single-zone constant-tariff case was identified as a Simple Linear Hybrid Automaton [13], establishing NP-hardness of the general problem and providing a theoretical anchor for RBRL as a near-optimal approximation.

Five principal findings emerge. First, RBRL reduces monthly per-zone cost by 18.2% (Riyadh) and 16.8% (Jeddah) in simulation relative to the unoptimised thermostat under the actual step-wise tariff, with zero comfort violations across all seven zone configurations and all four seasons. Second, cost model fidelity is as consequential as algorithmic choice: training under a linear tariff incurs a 13.1% true-cost penalty because the agent cannot perceive the Tier 1 pre-cooling threshold (Section 10.2 and Figure 2). Third, the hard rule layer is necessary for enforced comfort; the soft pre-cooling rule R3 adds a further 3.5% reduction. Fourth, adjacent zone temperatures—via $T_{iz,i}^{(t)}$ —must be in the agent state to prevent inter-zone cascades in grid configurations. Fifth, RBRL’s savings are stable at 17.0–18.6% from one zone to sixteen in Riyadh (Table 10), confirming scalability; Section 10.4 collects the full robustness evidence across seeds, topology, climate, season, thermal capacity, switching cost and evaluation horizon.

The practical recommendation is three-fold: (i) use the actual step-wise tariff in the optimisation objective—a simplified linear tariff incurs a 13.1% true-cost penalty; (ii) include the effective inter-zone temperature in the controller state; and (iii) in humid coastal climates such as Jeddah (SHR $\approx$ 0.65), future implementations should incorporate humidity-aware cost accounting, since the latent-heat fraction adds $\approx$ 54% to actual HVAC demand and a dehumidification-aware controller could time-shift latent loads to cheaper tariff tiers. All three are low-cost choices requiring no hardware changes. Future work will extend to latent-heat-aware RC models [45], stochastic occupancy, rooftop PV integration [19], federated multi-building control [32], [42], and hardware-in-the-loop validation.

12. CRediT Author Statement

The corresponding author: Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualisation.

13. Declaration of Competing Interest

The author declares no conflicts of interest.

14. Ethics Declaration

This study is based entirely on computational simulation using publicly available weather data (Energy-Plus Weather files) and standardised building parameters (Saudi Building Code 2018, ASHRAE 55-2023). No human subjects, animals, or personal data were involved in any stage of this research. No ethical approval was required for this work.

15. Data Availability

The run configurations for both cities, the weather-file specifications, and a reference implementation of the RBRL framework are available at: [Hybrid Rule-Constrained PPO for Cost-Effective Binary HVAC Scheduling in Saudi Residential Buildings](#). The weather data are public: the IWEC typical-year files for Riyadh (WMO 404380) and Jeddah (WMO 410240) may be downloaded from the EnergyPlus weather repository [46] and are not redistributed here. All building parameters are those of Table 2 and follow SBC 2018, with the comfort band taken from ASHRAE 55-2023 [40], so the model is fully specified by the manuscript.

References

- [1] L. Pérez-Lombard, J. Ortiz, C. Pout, A review on buildings energy consumption information, *Energy and Buildings* 40 (3) (2008) 394–398. doi:10.1016/j.enbuild.2007.03.007.
- [2] M. S. Al-Homoud, M. Krarti, Energy efficiency of residential buildings in the Kingdom of Saudi Arabia: Review of status and future roadmap, *Journal of Building Engineering* 36 (2021) 102143. doi:10.1016/j.jobbe.2020.102143.
- [3] M. Krarti, M. Aldubyan, E. Williams, Residential building stock model for evaluating energy retrofit programs in Saudi Arabia, *Energy* 195 (2020) 116980. doi:10.1016/j.energy.2020.116980.
- [4] F. Belaïd, M. G. Fikru, Resolving the energy efficiency paradox: leveraging benefits for Saudi Arabia’s building sector, *Humanities and Social Sciences Communications* 12 (2025) 1573. doi:10.1057/s41599-025-05078-2.
- [5] K. M. Nahiduzzaman, A. S. Abdallah, A. Moradzadeh, A. M. Shotorbani, K. Hewage, R. Sadiq, Impacts of tariffs on energy conscious behavior with respect to household attributes in Saudi Arabia, *Energies* 16 (3) (2023) 1458. doi:10.3390/en16031458.
- [6] Saudi Electricity Regulatory Authority (SERA), Saudi Electricity Company (SEC), Residential consumption tariff, <https://www.sera.gov.sa/en/consumer/electric-tariff/electric-tariff-categories/consumption-tariff>, post-2018 two-block residential schedule: 0.18 SAR/kWh up to 6,000 kWh/month and 0.30 SAR/kWh above; fixed meter charge and 15% VAT apply. Accessed 2026. (2024).
- [7] M. J. Risbeck, C. T. Maravelias, J. B. Rawlings, R. D. Turney, A mixed-integer linear programming model for real-time cost optimization of building heating, ventilation, and air conditioning equipment, *Energy and Buildings* 142 (2017) 220–235. doi:10.1016/j.enbuild.2017.02.053.
- [8] K. Al Sayed, A. Boodi, R. S. Broujeny, K. Beddiar, Reinforcement learning for HVAC control in intelligent buildings: a technical and conceptual review, *Journal of Building Engineering* 95 (2024) 110085. doi:10.1016/j.jobbe.2024.110085.
- [9] M. Biemann, F. Scheller, X. Liu, L. Huang, Experimental evaluation of model-free reinforcement learning algorithms for continuous HVAC control, *Applied Energy* 298 (2021) 117164. doi:10.1016/j.apenergy.2021.117164.

- [10] D. Azuatalam, W.-L. Lee, F. de Nijs, T. Liddle, Reinforcement learning for whole-building HVAC control and demand response, *Energy and AI* 2 (2020) 100020. doi:[10.1016/j.egyai.2020.100020](https://doi.org/10.1016/j.egyai.2020.100020).
- [11] S. Xu, Y. Fu, Y. Wang, Z. Yang, C. Huang, Z. O'Neill, Z. Wang, Q. Zhu, Efficient and assured reinforcement learning-based building HVAC control with heterogeneous expert-guided training, *Scientific Reports* 15 (2025) 7677. doi:[10.1038/s41598-025-91326-z](https://doi.org/10.1038/s41598-025-91326-z).
- [12] F. M. Solinas, A. Macii, E. Patti, L. Bottaccioli, An online reinforcement learning approach for HVAC control, *Expert Systems with Applications* 238 (2024) 121749. doi:[10.1016/j.eswa.2023.121749](https://doi.org/10.1016/j.eswa.2023.121749).
- [13] M. A. A. Mousa, S. Schewe, D. Wojtczak, Optimal control for simple linear hybrid systems, in: *Proceedings of the 23rd International Symposium on Temporal Representation and Reasoning (TIME)*, IEEE, 2016, pp. 12–20. doi:[10.1109/TIME.2016.9](https://doi.org/10.1109/TIME.2016.9).
- [14] P. Bacher, H. Madsen, Identifying suitable models for the heat dynamics of buildings, *Energy and Buildings* 43 (7) (2011) 1511–1522. doi:[10.1016/j.enbuild.2011.02.005](https://doi.org/10.1016/j.enbuild.2011.02.005).
- [15] M. A. A. Mousa, S. Schewe, D. Wojtczak, Optimal control for multi-mode systems with discrete costs, in: *Formal Modeling and Analysis of Timed Systems (FORMATS 2017)*, Vol. 10419 of *Lecture Notes in Computer Science*, Springer, 2017, pp. 77–96. doi:[10.1007/978-3-319-65765-3_5](https://doi.org/10.1007/978-3-319-65765-3_5).
- [16] A. Al-Reshidi, N. Al-Tamimi, Z. Al-Otaibi, M. Al-Hamdan, N. Alajlan, Energy performance analysis of residential buildings in Saudi Arabia under different climatic zones and code compliance scenarios, *Energy and Buildings* 312 (2024) 114199. doi:[10.1016/j.enbuild.2024.114199](https://doi.org/10.1016/j.enbuild.2024.114199).
- [17] E. Alayed, R. O'Hegarty, O. Kinnane, Thermal envelope analysis for new code compliance of Saudi Arabian dwellings, *Energy and Buildings* 243 (2021) 110997. doi:[10.1016/j.enbuild.2021.110997](https://doi.org/10.1016/j.enbuild.2021.110997).
- [18] F. Belaïd, J. I. Mikayilov, Closing the efficiency gap: insights into curbing the direct rebound effect of residential electricity consumption in Saudi Arabia, *Energy Economics* 135 (2024) 107647. doi:[10.1016/j.eneco.2024.107647](https://doi.org/10.1016/j.eneco.2024.107647).
- [19] A. M. Alotaibi, T. K. Makhdoom, A. B. S. Alquaity, Pathways toward improving the energy efficiency of residential air-conditioning systems in Saudi Arabia, *ASME Journal of Solar Energy Engineering* 146 (5) (2024) 051010. doi:[10.1115/1.4065973](https://doi.org/10.1115/1.4065973).
- [20] H. T. Dinh, J. Cho, D. Kim, MILP-based optimal day-ahead scheduling for a system-centric community energy management system supporting different types of homes and energy trading, *Scientific Reports* 12 (2022) 18305. doi:[10.1038/s41598-022-22293-y](https://doi.org/10.1038/s41598-022-22293-y).
- [21] J. Drgoňa, J. Arroyo, I. C. Figueroa, D. Blum, K. Arendt, D. Kim, E. P. Ollé, J. Oravec, M. Wetter, D. L. Vrabie, L. Helsen, All you need to know about model predictive control for buildings, *Annual Reviews in Control* 50 (2020) 190–232. doi:[10.1016/j.arcontrol.2020.09.001](https://doi.org/10.1016/j.arcontrol.2020.09.001).
- [22] Y. Du, H. Zandi, O. Kotevska, K. Kurte, J. Munk, K. Amasyali, E. McKee, F. Li, Intelligent multi-zone residential HVAC control strategy based on deep reinforcement learning, *Applied Energy* 281 (2021) 116117. doi:[10.1016/j.apenergy.2020.116117](https://doi.org/10.1016/j.apenergy.2020.116117).
- [23] Y. Du, F. Li, J. Munk, K. Kurte, O. Kotevska, K. Amasyali, H. Zandi, Multi-task deep reinforcement learning for intelligent multi-zone residential HVAC control, *Electric Power Systems Research* 192 (2021) 106959. doi:[10.1016/j.epsr.2020.106959](https://doi.org/10.1016/j.epsr.2020.106959).
- [24] Z. Li, W. Zhang, Y. Chen, Y. Ding, Multi-zone HVAC control with inter-zone thermal coupling: a deep reinforcement learning approach, *Applied Energy* 356 (2024) 122416. doi:[10.1016/j.apenergy.2023.122416](https://doi.org/10.1016/j.apenergy.2023.122416).

- [25] X. Ding, W. Du, A. Cerpa, Multi-zone HVAC control with model-based deep reinforcement learning, *IEEE Transactions on Automation Science and Engineering* 21 (4) (2024) 4406–4418. [doi:10.1109/TASE.2024.3410951](#).
- [26] S. Liu, Z. Chen, Q. Li, B. Chen, Enhancing HVAC energy management through multi-zone occupant-centric approach: a multi-agent deep reinforcement learning solution, *Energy and Buildings* 303 (2024) 113770. [doi:10.1016/j.enbuild.2023.113770](#).
- [27] F. Hou, J. Ma, H.-L. Kwok, J. C. P. Cheng, Multi-source transfer learning method for enhancing the deployment of deep reinforcement learning in multi-zone building HVAC control, *Energy and Buildings* 314 (2024) 114608. [doi:10.1016/j.enbuild.2024.114608](#).
- [28] A. Manjavacas, A. Campoy-Nieves, J. Jiménez-Raboso, M. Molina-Solana, J. Gómez-Romero, An experimental evaluation of deep reinforcement learning algorithms for HVAC control, *Artificial Intelligence Review* 57 (7) (2024) 173. [doi:10.1007/s10462-024-10819-x](#).
- [29] H. Wang, X. Chen, N. Vital, E. Duffy, A. Razi, Energy optimization for HVAC systems in multi-VAV open offices: a deep reinforcement learning approach, *Applied Energy* 356 (2024) 122354. [doi:10.1016/j.apenergy.2023.122354](#).
- [30] Y. Guo, Z. Tan, H. Chen, G. Li, J. Wang, R. Huang, J. Liu, T. Ahmad, Deep reinforcement learning-based approach for optimizing energy conversion in a distributed multi-energy system considering CO₂ capture, wind power and storage, *Applied Energy* 368 (2025) 124467. [doi:10.1016/j.apenergy.2024.124467](#).
- [31] S. Faddel, Q. Sun, G. Tian, A. Parlato, Scheduling of the HVAC system in a real commercial building considering equipment cycling and rebound effects, *Frontiers in Energy Research* 11 (2023) 1283369. [doi:10.3389/fenrg.2023.1283369](#).
- [32] W. Xue, N. Jia, M. Zhao, Multi-agent deep reinforcement learning based HVAC control for multi-zone buildings considering zone-energy-allocation optimization, *Energy and Buildings* 329 (2025) 115241. [doi:10.1016/j.enbuild.2024.115241](#).
- [33] R. El Makroum, A. Khallaayoun, R. Lghoul, K. Mehta, W. Zörner, Home energy management system based on genetic algorithm for load scheduling: a case study based on real life consumption data, *Energies* 16 (6) (2023) 2698. [doi:10.3390/en16062698](#).
- [34] I. Gonçalves, A. Gomes, C. Henggeler Antunes, A comprehensive and modular set of appliance operation MILP models for demand response optimization, *Applied Energy* 318 (2022) 119140. [doi:10.1016/j.apenergy.2022.119140](#).
- [35] J. Drgoňa, A. R. Tuor, V. Chandan, D. L. Vrabie, Physics-constrained deep learning of multi-zone building thermal dynamics, *Energy and Buildings* 243 (2021) 110992. [doi:10.1016/j.enbuild.2021.110992](#).
- [36] J. Wang, Y. Jiang, C.-Y. Tang, L. Song, Development and validation of a second-order thermal network model for residential buildings, *Applied Energy* 306 (2021) 118124. [doi:10.1016/j.apenergy.2021.118124](#).
- [37] Y. Deng, S. Cao, N. Chen, X. Li, J. Zhang, An RC-network approach for HVAC precooling optimization in buildings, *IEEE Transactions on Sustainable Computing* 7 (3) (2022) 726–739. [doi:10.1109/TSUSC.2019.2945805](#).
- [38] L. Di Natale, B. Svetozarevic, P. Heer, C. N. Jones, Towards scalable physically consistent neural networks: an application to data-driven multi-zone thermal building models, *Applied Energy* 340 (2023) 121071. [doi:10.1016/j.apenergy.2023.121071](#).

- [39] Saudi Building Code National Committee, Saudi building code (SBC) 2018: Implementation and enforcement, <https://sbc.gov.sa>, sBC compliance is mandatory for all construction in the Kingdom and is enforced through municipal permitting and inspection; the completion/occupancy certificate required for utility connection is contingent on compliance. Council of Ministers Decision No. 286 (9 Oct 2024) amended Article 9 of the SBC Implementation Law to require the occupancy certificate for full activation of the electricity connection. (2024).
- [40] ASHRAE, Thermal environmental conditions for human occupancy: ANSI/ASHRAE standard 55-2023, aNSI-approved standard; no DOI assigned. Available: <https://www.ashrae.org/technical-resources/bookstore/standard-55> (2023).
- [41] V. Zamani, S. Abtahi, Y. Chen, Y. Li, Parameter-input estimation of RC thermal models of buildings using unscented Kalman filter and nonlinear least square method, *Indoor and Built Environment* 34 (1) (2024) 41–75. doi:10.1177/1420326X241270775.
- [42] L. Yu, Y. Sun, Z. Xu, C. Shen, D. Yue, T. Jiang, X. Guan, Multi-agent deep reinforcement learning for HVAC control in commercial buildings, *IEEE Transactions on Smart Grid* 12 (1) (2021) 407–419. doi:10.1109/TSG.2020.3011739.
- [43] E. P. Bandarra Filho, G. Martins, M. B. Riaz, S. M. Bilal, O. S. H. Mendonza, Optimization of air conditioning mechanical ventilation using simulated annealing for enhanced energy efficiency and cost reduction, *Scientific Reports* 15 (2025) 23429. doi:10.1038/s41598-025-07640-z.
- [44] S. Haddad, W. Zhang, R. Paolini, K. Gao, M. Altheeb, A. Al Mogirah, A. Bin Moammar, T. Hong, A. Khan, C. Cartalis, A. Polydoros, M. Santamouris, Quantifying the energy impact of heat mitigation technologies at the urban scale, *Nature Cities* 1 (2024) 62–72. doi:10.1038/s44284-023-00005-5.
- [45] L. Rodrigues, B. A. Cherian, S. Tokbolat, Reducing cooling energy demand in Saudi Arabian residential buildings using passive design approaches, *Buildings* 15 (11) (2025) 1895. doi:10.3390/buildings15111895.
- [46] U.S. Department of Energy, EnergyPlus weather data documentation, Tech. rep., U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, typical-year (IWECC) files used in this study: SAU_Riyadh.404380_IWECC.epw (WMO 404380) and SAU_Jeddah.410240_IWECC.epw (WMO 410240). No DOI assigned. Available: <https://energyplus.net/weather> (2023).
- [47] D. Bienvenido-Huertas, D. Sánchez-García, B. Tejedor, C. Rubio-Bellido, Energy savings in buildings applying ASHRAE 55 and regional adaptive thermal comfort models, *Urban Climate* 55 (2024) 101892. doi:10.1016/j.uclim.2024.101892.
- [48] P. Lamsal, S. B. Bajracharya, H. B. Rijal, A review on adaptive thermal comfort of office building for energy-saving building design, *Energies* 16 (3) (2023) 1524. doi:10.3390/en16031524.
- [49] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann, Stable-Baselines3: Reliable reinforcement learning implementations, *Journal of Machine Learning Research* 22 (268) (2021) 1–8. doi:10.5281/zenodo.5765968.